

EPILEPSY DIGITAL TWIN REPORT • INTEGRATED NEUROLOGICAL DECISION SUPPORT

Neuro Digital Twin Integrated Clinical Report

An integrated clinical assessment combining follow-up visits, seizure history, EEG, ECG, CT/MRI imaging, laboratory results, and pharmacogenomic testing for Focal impaired-awareness seizures.

Patient & Clinical Context		11/11 data categories included · generated 2026-09-23T16:21:26				
PATIENT John Doe Patient ID SYN-100482	AGE / SEX 34 years · Male	PRIMARY DIAGNOSIS Focal impaired-awareness seizures Secondary diagnosis: Secondary generalized tonic-clonic seizures	DIAGNOSED SINCE 12 May 2021			
CURRENT CLINICAL STATUS Stable outpatient epilepsy state with persistent breakthrough seizure risk						
CURRENT THERAPY Levetiracetam (Keppra) 1000 mg BID · Lamotrigine 150 mg BID						
REPORTING PERIOD Data reviewed: 01 Jan – 30 Jun 2026 (~6 months) Clinic visit notes cover 01 Jan 2026 to 02 Mar 2026; laboratory, imaging, EEG/ECG, and questionnaire data continue through 30 Jun 2026.						
5	12	6	6	12	35	12
Clinical Visits	CBC Reports	CT Studies	MRI Series	ECG Sessions	EEG Recordings	Questionnaires

01 Executive Summary

Across five clinical-note visits from 2026-01-01 to 2026-03-02, the patient has chronic focal epilepsy with persistent breakthrough seizures on levetiracetam and lamotrigine. The clinician repeatedly documented reasonably stable seizure control and favorable quality-of-life trajectory, while seizure burden, sleep quality, trigger exposure, aura symptoms, and medication tolerability continued to fluctuate.

67% Overall Health Score	Moderate Overall Clinical Risk	Elevated EEG-Derived Seizure Risk	10 (7 high) Genetic Safety Flags
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Domain the patient rates worst on questionnaires: Epileptic Triggers. See the Digital Twin Risk Dashboard (Section 13) for the full risk breakdown.

02 6-Month Timeline

- 01 JAN 2026

Routine epilepsy follow-up with stable clinician impression — Contributes to longitudinal assessment of seizure burden, triggers, sleep, and tolerability.
- 16 JAN 2026

Seizure clustering documented during follow-up — Raises near-term seizure-monitoring priority despite otherwise stable clinician impression.
- 03 FEB 2026

CBC/LFT safety review documented — Supports medication-safety monitoring without a documented abnormal lab signal.
- 15 FEB 2026

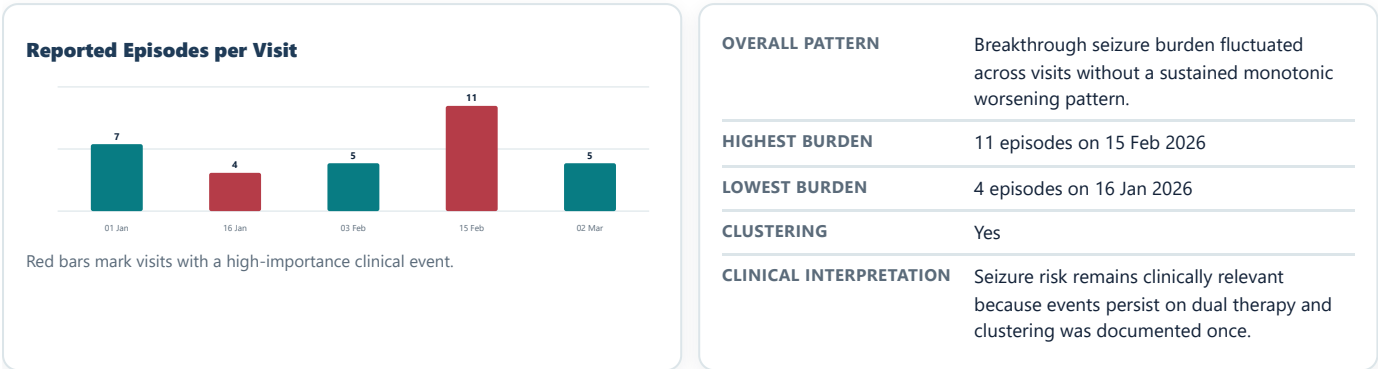
Highest recorded seizure burden in clinical-note window — 11 reported episodes make this the peak burden visit.
- 02 MAR 2026

Routine epilepsy follow-up with stable clinician impression — Contributes to longitudinal assessment of seizure burden, triggers, sleep, and tolerability.
- 10 SEP 2026

Pharmacogenomic (PGx) panel reported — 76 variants analyzed across 47 medications; see Pharmacogenomics.

03 Seizure Trends

Clinical-note episode counts, clustering, and duration



05 Medication Response

Current regimen, tolerability, and reported side effects

Current Regimen

Drug	Dose
Levetiracetam (Keppra)	1000 mg BID
Lamotrigine	150 mg BID

Effectiveness	Partial response
Tolerability	Generally tolerated with intermittent dizziness and variable side-effect burden
Documented Toxicity	No
Clinical Interpretation	Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.

Persistent Side Effects (clinical notes)

Memory fog post-dose

Tremor in hands

Weight change since last visit

Gum hypertrophy

Mood irritability

Nausea after taking medication

Skin rash or itching

Daytime drowsiness

Dizziness/lightheadedness

Unsteady gait (ataxia)

Medication Side Effects — Questionnaire Domain

Latest score **78.3/100** (first 68.3, trend increasing)

- Rate the severity of daytime drowsiness or somnolence: **Mild (Manageable)**
- Experience of dizziness, vertigo, or lightheadedness after taking medication: **Never**
- Presence of physical instability, unsteadiness, or clumsy movements (ataxia): **Slight imbalance**
- Frequency of unprovoked irritability, anger, or mood volatility: **Mild irritability**
- Signs of dermatological distress or unexplained skin rashes: **Clear skin**

06 Drug Interactions

2 current medication(s) reconciled · Flockhart Cytochrome P450 Drug Interaction Table (curated-snapshot-2024)

DDI & Patient-Specific Therapy Assessment — Summary

Overall: 1 clinically relevant findings | 2 unresolved finding(s).

3
Total Findings Assessed

0
Clinically Significant

1
Resolved — No Concern

2
Unresolved

MEDICATION	FLAG	KEY FINDING	PATIENT IMPACT	EVIDENCE SOURCE
Lamotrigine	UNRESOLVED	Therapeutic evidence present, but clinical effectiveness remains incomplete.	Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.	Patient pharmacogenomic findings; Laboratory findings; Clinical record; EEG findings
Levetiracetam (Keppra)	UNRESOLVED	Therapeutic evidence present, but clinical effectiveness remains incomplete.	Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.	Patient pharmacogenomic findings; Laboratory findings; Clinical record; EEG findings

Key clinical interpretation:

These flags identify findings that require ongoing clinical monitoring, not an automatic medication change. Each finding should be interpreted alongside the patient's actual treatment response, adverse effects, and the clinical/laboratory findings documented elsewhere in this report -- a resolved or unresolved flag is not itself a prescribing instruction.

Monitoring / Clinician Review Checklist

Medication reconciliation (dose, frequency, and route consistent across all source records)

Actual treatment response (seizure/symptom control on the current regimen)

Relevant adverse effects reported since the last review

Therapeutic drug levels, when indicated and available

Relevant laboratory and clinical parameters (e.g. renal, hepatic, hematological)

Medication Reconciliation

DRUG	DOSE	FLAGS
Lamotrigine	150.0 mg BID	—
Levetiracetam (Keppra)	1000.0 mg BID	—

No conflicting doses found for the same drug across the reconciled source data.

Drug Interaction Flag Sheet — Extracted Drug List

Positive Impact = curated/literature evidence explicitly describes a favorable or well-tolerated combination; **Negative Impact** = a real curated pharmacokinetic mechanism, a pharmacodynamic-risk rule, or literature language explicitly describing a risk. A pair with no evidence, or whose evidence never clearly resolved either way, is not listed here at all -- it is never guessed into either column (see Known Limitations).

Screened 26 drug(s) across 325 unique pairs; 146 pair(s) resolved clearly to Positive or Negative and are shown below. 179 pair(s) had no evidence, or evidence that did not clearly resolve either way, and are not listed here -- never shown as a negative/no-concern finding (see Known Limitations).

ARIPIRAZOLE	
POSITIVE IMPACT	NEGATIVE IMPACT
Aripiprazole + Citalopram Findings: Study to Evaluate the Efficacy, Safety and Tolerability of an Oral Aripiprazole/Escitalopram Combination Therapy in Participants With Major Depressive Disorder (MDD) Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Aripiprazole + Carbamazepine Findings: Aripiprazole's DailyMed label states: "...Reduce the aripiprazole dosage when administered concomitantly with a strong CYP3A4 inhibitor or a strong CYP2D6 inhibitor [see Dosage and Administration (2.6)]. Strong CYP3A4 Inducers (e.g., carbamazepine, rifampin) Concomitant use of aripiprazole and carbamazepine decreased the exposure of aripiprazole compared to the use of aripiprazole alone [see Clinical Pharmacology (12.3)]...." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Aripiprazole + Duloxetine Findings: Effect of augmentation with aripiprazole or augmentation with repetitive transcranial magnetic stimulation versus switching to the antidepressant venlafaxine extended release/duloxetine on cognition: A comparative effectiveness research trial for antidepressant incomplete and non-responders with treatment-resistant depression (ASCERTAIN-TRD). Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed	Aripiprazole + Fluoxetine Findings: Aripiprazole's DailyMed label states: "...lly Important Interactions with Aripiprazole Table 22: Clinically Important Drug Interactions with Aripiprazole: Concomitant Drug Name or Drug Class Clinical Rationale Clinical Recommendation Strong CYP3A4 Inhibitors (e.g., itraconazole, clarithromycin) or strong CYP2D6 inhibitors (e.g., quinidine, fluoxetine, paroxetine) Concomitant use of aripiprazole with strong CYP3A4 or CYP2D6 inhibitors increased the exposure of aripiprazole compared to the use of aripiprazole alone [see Clinical Pharmacology (12.3)]...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily)
Aripiprazole + Escitalopram Findings: Study to Evaluate the Efficacy, Safety and Tolerability of an Oral Aripiprazole/Escitalopram Combination Therapy in Participants With Major Depressive Disorder (MDD)	

POSITIVE IMPACT	NEGATIVE IMPACT
Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Hyperlink: DailyMed
Aripiprazole + Lamotrigine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Aripiprazole + Olanzapine Findings: Pregnancy exposure to olanzapine, quetiapine, risperidone, aripiprazole and risk of congenital malformations. A systematic review. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
Aripiprazole + Lithium Findings: Comparative efficacy, tolerability and acceptability of intravenous racemic ketamine with intranasal esketamine, aripiprazole and lithium as augmentative treatments for treatment-resistant unipolar depression: A systematic review and network meta-analysis. Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed	Aripiprazole + Paroxetine Findings: Aripiprazole's DailyMed label states: "...t Interactions with Aripiprazole Table 22: Clinically Important Drug Interactions with Aripiprazole: Concomitant Drug Name or Drug Class Clinical Rationale Clinical Recommendation Strong CYP3A4 Inhibitors (e.g., itraconazole, clarithromycin) or strong CYP2D6 inhibitors (e.g., quinidine, fluoxetine, paroxetine) Concomitant use of aripiprazole with strong CYP3A4 or CYP2D6 inhibitors increased the exposure of aripiprazole compared to the use of aripiprazole alone [see Clinical Pharmacology (12.3)]...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Aripiprazole + Oxcarbazepine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Aripiprazole + Quetiapine Findings: Pregnancy exposure to olanzapine, quetiapine, risperidone, aripiprazole and risk of congenital malformations. A systematic review. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
Aripiprazole + Sertraline Findings: Aripiprazole/Sertraline Combination: Clinical and Cost-Effectiveness in Comparison With Quetiapine for the Treatment of Bipolar Depression (ASCEnD Trial)-Protocol for a Nested Qualitative Study. Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed	
Aripiprazole + Venlafaxine Findings: Comparative Effectiveness Research Trial for Antidepressant Incomplete and Nonresponders With Treatment Resistant Depression (ASCERTAIN-TRD): Effect of Aripiprazole or Repetitive Transcranial Magnetic Stimulation Augmentation Versus Switching to the Antidepressant Venlafaxine on Quality of Life. Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed	

ATOMOXETINE

POSITIVE IMPACT	NEGATIVE IMPACT
Atomoxetine + Valproic Acid Findings: Improvement by methylphenidate and atomoxetine of social interaction deficits and recognition memory impairment in a mouse model of valproic acid-induced autism. Evidence: External evidence (PubMed); General web search (Tavily) Hyperlink: PubMed	Atomoxetine + Atorvastatin Findings: Atorvastatin Interactions: Risks, Effects, & Recommendations Evidence: General web search (Tavily) Hyperlink: General web search (Tavily)
	Atomoxetine + Citalopram Findings: Unraveling the Nature of Impaired Pain Inhibition in Patients With Chronic Whiplash-associated Disorders Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov
	Atomoxetine + Methylphenidate Findings: Methylphenidate and Atomoxetine in Pregnancy and Possible Adverse Fetal Outcomes: A Systematic Review and Meta-Analysis. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
	Atomoxetine + Oxcarbazepine Findings: Acute oxcarbazepine and atomoxetine overdose with quetiapine. Evidence: External evidence (PubMed); General web search (Tavily) Hyperlink: PubMed
	Atomoxetine + Paroxetine Findings: Paroxetine's DailyMed label states: "...Intervention Decrease the dosage of a CYP2D6 substrate if needed with concomitant paroxetine use. Conversely, an increase in dosage of a CYP2D6 substrate may be needed if paroxetine is discontinued. Examples propafenone, flecainide, atomoxetine, desipramine,

POSITIVE IMPACT	NEGATIVE IMPACT
	dextromethorphan, metoprolol, nebivolol, perphenazine, tolterodine, venlafaxine, risperidone...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); General web search (Tavily) Hyperlink: DailyMed
	Atomoxetine + Quetiapine Findings: Acute oxcarbazepine and atomoxetine overdose with quetiapine. Evidence: External evidence (PubMed); General web search (Tavily) Hyperlink: PubMed
	Atomoxetine + Sertraline Findings: Sertraline's DailyMed label states: "...Intervention: Decrease the dosage of a CYP2D6 substrate if needed with concomitant sertraline hydrochloride use. Conversely, an increase in dosage of a CYP2D6 substrate may be needed if sertraline hydrochloride is discontinued. Examples: propafenone, flecainide, atomoxetine, desipramine, dextromethorphan, metoprolol, nebivolol, perphenazine, thioridazine, tolterodine, venlafaxine Phenytoin Clinical Impact: Phenytoin is a narrow therapeutic index drug. Sertraline hydrochloride may increase phenytoin concentrations...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed

ATORVASTATIN

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	Atorvastatin + Atomoxetine Findings: Atorvastatin Interactions: Risks, Effects, & Recommendations Evidence: General web search (Tavily) Hyperlink: General web search (Tavily)
	Atorvastatin + Clobazam Findings: Atorvastatin Interactions: Risks, Effects, & Recommendations Evidence: General web search (Tavily) Hyperlink: General web search (Tavily)
	Atorvastatin + Fluoxetine Findings: Atorvastatin and Fluoxetine Prevent Oxidative Stress and Mitochondrial Dysfunction Evoked by Glutamate Toxicity in Hippocampal Slices. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
	Atorvastatin + Quetiapine Findings: Adverse effects of interactions between antipsychotics and ... Evidence: General web search (Tavily) Hyperlink: General web search (Tavily)
	Atorvastatin + Venlafaxine Findings: Atorvastatin Interactions: Risks, Effects, & Recommendations Evidence: General web search (Tavily) Hyperlink: General web search (Tavily)

BUPROPION

POSITIVE IMPACT	NEGATIVE IMPACT
Bupropion + Olanzapine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Bupropion + Carbamazepine Findings: Bupropion's DailyMed label states: "(7.1) • Drugs metabolized by CYP2D6: Bupropion inhibits CYP2D6 and can increase concentrations of: antidepressants (e.g., venlafaxine, nortriptyline, imipramine, desipramine, paroxetine,..." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Bupropion + Oxcarbazepine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Bupropion + Escitalopram Findings: Risks of Mixing Lexapro and Wellbutrin (Bupropion) Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: General web search (Tavily)
Bupropion + Valproic Acid Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Bupropion + Fluoxetine Findings: Bupropion's DailyMed label states: "... (7.1) • Drugs metabolized by CYP2D6: Bupropion inhibits CYP2D6 and can increase concentrations of:

POSITIVE IMPACT	NEGATIVE IMPACT
	antidepressants (e.g., venlafaxine, nortriptyline, imipramine, desipramine, paroxetine, fluoxetine, sertraline), antipsychotics (e.g., haloperidol, risperidone, thioridazine), beta-blockers (e.g., metoprolol), and Type 1C antiarrhythmics (e.g., propafenone, flecainide). Consider dose reduction when using with bupropion...." Evidence: DailyMed label (Drug Interactions section); General web search (Tavily) Hyperlink: DailyMed
	Bupropion + Haloperidol Findings: Bupropion's DailyMed label states: "... (7.1) • Drugs metabolized by CYP2D6: Bupropion inhibits CYP2D6 and can increase concentrations of: antidepressants (e.g., venlafaxine, nortriptyline, imipramine, desipramine, paroxetine, fluoxetine, sertraline), antipsychotics (e.g., haloperidol, risperidone, thioridazine), beta-blockers (e.g., metoprolol), and Type 1C antiarrhythmics (e.g., propafenone, flecainide). Consider dose reduction when using with bupropion. (7.2) • Drugs that lower seizure threshold: Dose bupropion hydrochloride extended-release tablets (XL) with caution...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Bupropion + Lamotrigine Findings: Massive lamotrigine and bupropion overdose resulting in status epilepticus without cardiovascular collapse. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
	Bupropion + Nortriptyline Findings: Bupropion's DailyMed label states: "... (7.1) • Drugs metabolized by CYP2D6: Bupropion inhibits CYP2D6 and can increase concentrations of: antidepressants (e.g., venlafaxine, nortriptyline, imipramine, desipramine, paroxetine, fluoxetine, sertraline), antipsychotics (e.g., haloperidol, risperidone, thioridazine), beta-blockers (e.g., metoprolol), and Type 1C antiarrhythmics (e.g., propafenone, flecainide). Consider dose reduction when using with bupropion...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Bupropion + Paroxetine Findings: Bupropion's DailyMed label states: "... (7.1) • Drugs metabolized by CYP2D6: Bupropion inhibits CYP2D6 and can increase concentrations of: antidepressants (e.g., venlafaxine, nortriptyline, imipramine, desipramine, paroxetine, fluoxetine, sertraline), antipsychotics (e.g., haloperidol, risperidone, thioridazine), beta-blockers (e.g., metoprolol), and Type 1C antiarrhythmics (e.g., propafenone, flecainide). Consider dose reduction when using with bupropion...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Bupropion + Phenytoin Findings: Bupropion's DailyMed label states: "(7.1) • Drugs metabolized by CYP2D6: Bupropion inhibits CYP2D6 and can increase concentrations of: antidepressants (e.g., venlafaxine, nortriptyline, imipramine, desipramine, paroxetine, fluoxetine, sertraline), anti..." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Bupropion + Sertraline Findings: Bupropion's DailyMed label states: "... (7.1) • Drugs metabolized by CYP2D6: Bupropion inhibits CYP2D6 and can increase concentrations of: antidepressants (e.g., venlafaxine, nortriptyline, imipramine, desipramine, paroxetine, fluoxetine, sertraline), antipsychotics (e.g., haloperidol, risperidone, thioridazine), beta-blockers (e.g., metoprolol), and Type 1C antiarrhythmics (e.g., propafenone, flecainide). Consider dose reduction when using with bupropion...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Bupropion + Venlafaxine Findings: Bupropion's DailyMed label states: "... (7.1) • Drugs metabolized by CYP2D6: Bupropion inhibits CYP2D6 and can increase concentrations of: antidepressants (e.g., venlafaxine, nortriptyline, imipramine, desipramine, paroxetine, fluoxetine, sertraline), antipsychotics (e.g., haloperidol, risperidone, thioridazine),

POSITIVE IMPACT	NEGATIVE IMPACT
	beta-blockers (e.g., metoprolol), and Type 1C antiarrhythmics (e.g., propafenone, flecainide). Consider dose reduction when using with bupropion...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed

CARBAMAZEPINE

POSITIVE IMPACT	NEGATIVE IMPACT
Carbamazepine + Citalopram Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Carbamazepine + Aripiprazole Findings: Aripiprazole's DailyMed label states: "...Reduce the aripiprazole dosage when administered concomitantly with a strong CYP3A4 inhibitor or a strong CYP2D6 inhibitor [see Dosage and Administration (2.6)]. Strong CYP3A4 Inducers (e.g., carbamazepine, rifampin) Concomitant use of aripiprazole and carbamazepine decreased the exposure of aripiprazole compared to the use of aripiprazole alone [see Clinical Pharmacology (12.3)]...." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Carbamazepine + Duloxetine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Carbamazepine + Bupropion Findings: Bupropion's DailyMed label states: "(7.1) • Drugs metabolized by CYP2D6: Bupropion inhibits CYP2D6 and can increase concentrations of: antidepressants (e.g., venlafaxine, nortriptyline, imipramine, desipramine, paroxetine,..." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Carbamazepine + Haloperidol Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Carbamazepine + Escitalopram Findings: Escitalopram's DailyMed label states: "...Intervention: If concomitant treatment with sumatriptan and an SSRI is clinically warranted, appropriate observation of the patient is advised [see Warning and Precautions (5.2)]. Carbamazepine Clinical Impact: Combined administration of racemic citalopram (40 mg/day for 14 days) and carbamazepine (titrated to 400 mg/day for 35 days) did not significantly affect the pharmacokinetics of carbamazepine, a CYP3A4 substrate...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Carbamazepine + Oxcarbazepine Findings: Oxcarbazepine's DailyMed label states: "(2.5 , 7.2) Oral Contraceptives : Advise patients that Oxtellar XR may decrease the effectiveness of hormonal contraceptives...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); General web search (Tavily) Hyperlink: DailyMed	Carbamazepine + Ethosuximide Findings: Carbamazepine Toxicity: Practice Essentials ... Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: General web search (Tavily)
Carbamazepine + Paroxetine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Carbamazepine + Fluoxetine Findings: Fluoxetine's DailyMed label states: "... (7.2) Benzodiazepines: Diazepam – increased t ½ , alprazolam - further psychomotor performance decrement due to increased levels. (7.7) Antipsychotics: Potential for elevation of haloperidol and clozapine levels. (7.7) Anticonvulsants: Potential for elevated phenytoin and carbamazepine levels and clinical anticonvulsant toxicity. (7.7) Serotonergic Drugs: (2.9 , 2.10 , 4.1 , 5.2) Drugs that Interfere with Hemostasis (e.g., NSAIDs, Aspirin, Warfarin): May potentiate the risk of bleeding...." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Carbamazepine + Perampanel Findings: Perampanel's DailyMed label states: "7 DRUG INTERACTIONS Contraceptives: 12 mg once daily may decrease the effectiveness of hormonal contraceptives containing levonorgestrel (7.1) Moderate and Strong CYP3A4 Inducers (including carbamazepine, oxcarbazepine, and phenytoin): increase clearance of perampanel and decrease perampanel plasma concentrations. When moderate or strong CYP3A4 inducers are introduced or withdrawn, monitor patients closely...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed	Carbamazepine + Lamotrigine Findings: Lamotrigine's DailyMed label states: "...preparations containing 30 mcg ethinylestradiol and 150 mcg levonorgestrel ↓ levonorgestrel Decrease in levonorgestrel component by 19%. Carbamazepine and carbamazepine epoxide ↓ lamotrigine Addition of carbamazepine decreases lamotrigine concentration approximately 40%. ? carbamazepine epoxide May increase carbamazepine epoxide levels. Lopinavir/ritonavir ↓ lamotrigine Decreased lamotrigine concentration approximately 50%...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Carbamazepine + Sertraline Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Carbamazepine + Levetiracetam Findings: Use of Phenytoin, Phenobarbital Carbamazepine, Levetiracetam Lamotrigine and Valproate in Pregnancy and Breastfeeding: Risk of Major
Carbamazepine + Venlafaxine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	

POSITIVE IMPACT	NEGATIVE IMPACT
	Malformations, Dose-dependency, Monotherapy vs Polytherapy, Pharmacokinetics and Clinical Implications. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
	Carbamazepine + Lithium Findings: Lithium's DailyMed label states: "...Intervention: More frequent serum lithium concentration monitoring. Increase lithium dosage based on serum lithium concentration and clinical response [see Dosage and Administration (2.3)] . Methylodopa, Phenytoin and Carbamazepine Clinical Impact: Concomitant use may increase risk of adverse reactions of these drugs. Intervention: Monitor patients closely for adverse reactions of methylodopa, phenytoin, and carbamazepine. Iodide Preparations Clinical Impact: Concomitant use may produce hypothyroidism...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Carbamazepine + Olanzapine Findings: Olanzapine's DailyMed label states: "Diazepam: May potentiate orthostatic hypotension (7.1, 7.2) Alcohol: May potentiate orthostatic hypotension (7.1) Carbamazepine: Increased clearance of olanzapine (7.1) Fluvoxamine: May increase olanzapine levels (7.1) Olanzapine and Fluoxetine in Combination: Also refer to the DrugInteractions section of the package insert for Symbyax...." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Carbamazepine + Phenytoin Findings: Carbamazepine's DailyMed label states: "...Agents that are CYP inducers that have been found, or are expected, to decrease plasma levels of carbamazepine extended-release capsules include, for example, the following: Aminophylline, cisplatin, doxorubicin HCL, felbamate, fosphenytoin, methsuximide, phenobarbital, phenytoin(1), primidone, rifampin and theophylline. (1) Phenytoin plasma levels have also been reported to increase and decrease in the presence of carbamazepine, see below...." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Carbamazepine + Quetiapine Findings: Quetiapine's DailyMed label states: "...Quetiapine exposure is increased by the prototype CYP3A4 inhibitors (e.g., ketoconazole, itraconazole, indinavir, ritonavir, nefazodone, etc.) and decreased by the prototype CYP3A4 inducers (e.g., phenytoin, carbamazepine, rifampin, avasimibe, St. John's wort etc.). Dose adjustment of quetiapine will be necessary if it is co-administered with potent CYP3A4 inducers or inhibitors...." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Carbamazepine + Valproic Acid Findings: Carbamazepine's DailyMed label states: "...Accordingly, the dosage of carbamazepine extended-release capsules should be adjusted and/or the plasma levels monitored when used concomitantly with loxapine, quetiapine, or valproic acid. Thus, if a patient has been titrated to a stable dosage of carbamazepine extended-release capsules, and then begins a course of treatment with one of these CYP3A4 or epoxide hydrolase inhibitors, it is reasonable to expect that a dose reduction for carbamazepine extended-release capsules may be ne..." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed

CITALOPRAM

POSITIVE IMPACT	NEGATIVE IMPACT
Citalopram + Aripiprazole Findings: Study to Evaluate the Efficacy, Safety and Tolerability of an Oral Aripiprazole/Escitalopram Combination Therapy in Participants With Major Depressive Disorder (MDD) Evidence: ClinicalTrials.gov; General web search (Tavily)	Citalopram + Atomoxetine Findings: Unraveling the Nature of Impaired Pain Inhibition in Patients With Chronic Whiplash-associated Disorders Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov

POSITIVE IMPACT	NEGATIVE IMPACT
Hyperlink: ClinicalTrials.gov Citalopram + Carbamazepine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov Citalopram + Haloperidol Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov Citalopram + Oxcarbazepine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov Citalopram + Valproic Acid Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Citalopram + Escitalopram Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Citalopram + Fluoxetine Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Citalopram + Lithium Findings: Citalopram's DailyMed label states: "...Other Serotonergic Drugs Clinical Impact: Concomitant use of citalopram and other serotonergic drugs (including other SSRIs, SNRIs, triptans, tricyclic antidepressants, opioids, lithium, buspirone, amphetamines, tryptophan, and St. John's Wort) increases the risk of serotonin syndrome. Intervention: Monitor patients for signs and symptoms of serotonin syndrome, particularly during citalopram initiation and dosage increases...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed Citalopram + Paroxetine Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; ClinicalTrials.gov; General web search (Tavily) Citalopram + Sertraline Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)

CLOBAZAM

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	Clobazam + Atorvastatin Findings: Atorvastatin Interactions: Risks, Effects, & Recommendations Evidence: General web search (Tavily) Hyperlink: General web search (Tavily) Clobazam + Quetiapine Findings: Central Pontine Myelinolysis: A Rare and Life-Threatening Adverse Effect of Clobazam and Quetiapine. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed

DULOXETINE

POSITIVE IMPACT	NEGATIVE IMPACT
Duloxetine + Aripiprazole Findings: Effect of augmentation with aripiprazole or augmentation with repetitive transcranial magnetic stimulation versus switching to the antidepressant venlafaxine extended release/duloxetine on cognition: A comparative effectiveness research trial for antidepressant incomplete and non-responders with treatment-resistant depression (ASCERTAIN-TRD). Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed Duloxetine + Carbamazepine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov Duloxetine + Haloperidol Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov Duloxetine + Lamotrigine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily)	Duloxetine + Fluoxetine Findings: Duloxetine's DailyMed label states: "...Similar effects would be expected with other potent CYP2D6 inhibitors (e.g., fluoxetine, quinidine) [see Warnings and Precautions (5.12)]" Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); General web search (Tavily) Hyperlink: DailyMed Duloxetine + Paroxetine Findings: Duloxetine's DailyMed label states: "...Other drugs that inhibit CYP1A2 metabolism include cimetidine and quinolone antimicrobials such as ciprofloxacin and enoxacin [see Warnings and Precautions (5.12)] . 7.2 Inhibitors of CYP2D6 Concomitant use of duloxetine (40 mg once daily) with paroxetine (20 mg once daily) increased the concentration of duloxetine AUC by about 60%, and greater degrees of inhibition are expected with higher doses of paroxetine...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed Duloxetine + Sertraline Findings: Duloxetine and Sertraline Interaction: Risks and Management empathia.ai Evidence: ClinicalTrials.gov; General web search (Tavily)

POSITIVE IMPACT	NEGATIVE IMPACT
Hyperlink: ClinicalTrials.gov	Hyperlink: General web search (Tavily)
Duloxetine + Lithium	
Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies	
Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)	
Hyperlink: ClinicalTrials.gov	
Duloxetine + Olanzapine	
Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies	
Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)	
Hyperlink: ClinicalTrials.gov	
Duloxetine + Oxcarbazepine	
Findings: Comparative efficacy and safety of gabapentin, pregabalin, oxcarbazepine, and duloxetine in diabetic peripheral neuropathy: A network meta-analysis.	
Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)	
Hyperlink: PubMed	
Duloxetine + Quetiapine	
Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies	
Evidence: ClinicalTrials.gov; General web search (Tavily)	
Hyperlink: ClinicalTrials.gov	
Duloxetine + Valproic Acid	
Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies	
Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)	
Hyperlink: ClinicalTrials.gov	

ESCITALOPRAM

POSITIVE IMPACT	NEGATIVE IMPACT
Escitalopram + Aripiprazole Findings: Study to Evaluate the Efficacy, Safety and Tolerability of an Oral Aripiprazole/Escitalopram Combination Therapy in Participants With Major Depressive Disorder (MDD) Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Escitalopram + Bupropion Findings: Risks of Mixing Lexapro and Wellbutrin (Bupropion) Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: General web search (Tavily)
Escitalopram + Haloperidol Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Escitalopram + Carbamazepine Findings: Escitalopram's DailyMed label states: "...Intervention: If concomitant treatment with sumatriptan and an SSRI is clinically warranted, appropriate observation of the patient is advised [see Warning and Precautions (5.2)]. Carbamazepine Clinical Impact: Combined administration of racemic citalopram (40 mg/day for 14 days) and carbamazepine (titrated to 400 mg/day for 35 days) did not significantly affect the pharmacokinetics of carbamazepine, a CYP3A4 substrate...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Escitalopram + Nortriptyline Findings: Effectiveness of Escitalopram and Nortriptyline on Depressive Symptoms in Parkinson's disease: the ADePT-PD RCT pilot. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed	Escitalopram + Citalopram Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)
Escitalopram + Olanzapine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Escitalopram + Fluoxetine Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; External evidence (PubMed); General web search (Tavily)
Escitalopram + Oxcarbazepine Findings: Comparison of clinical efficacy of oxcarbazepine and lamotrigine combined with escitalopram, and impact on prognostic quality of life in treating patients with epilepsy and depressive disorder. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed	Escitalopram + Lithium Findings: Escitalopram's DailyMed label states: "...Intervention: Escitalopram tablets is contraindicated in patients taking pimozide [see Contraindications (4)]. Serotonergic Drugs Clinical Impact: Concomitant use of Escitalopram tablets and other serotonergic drugs (including other SSRIs, SNRIs, triptans, tricyclic antidepressants, opioids, lithium, buspirone, amphetamines, tryptophan, and St. John's Wot) increases the risk of serotonin syndrome. Intervention: Monitor patients for signs and symptoms of serotonin syndrome, particularly during Escitalopram tablets initiation and dosage increases...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Escitalopram + Valproic Acid Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	

POSITIVE IMPACT	NEGATIVE IMPACT
	Evidence: Pharmacodynamic interaction reference; External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)
	Escitalopram + Sertraline Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)

ETHOSUXIMIDE

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	Ethosuximide + Carbamazepine Findings: Carbamazepine Toxicity: Practice Essentials ... Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: General web search (Tavily)
	Ethosuximide + Phenytoin Findings: Ethosuximide's DailyMed label states: "Drug Interactions Since ethosuximide may interact with concurrently administered antiepileptic drugs, periodic serum level determinations of these drugs may be necessary (e.g., ethosuximide may elevate phenytoin serum levels and valproic acid has been reported to both increase and decrease ethosuximide levels)." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Ethosuximide + Valproic Acid Findings: Ethosuximide's DailyMed label states: "Drug Interactions Since ethosuximide may interact with concurrently administered antiepileptic drugs, periodic serum level determinations of these drugs may be necessary (e.g., ethosuximide may elevate phenytoin serum levels and valproic acid has been reported to both increase and decrease ethosuximide levels)." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed

FLUOXETINE

POSITIVE IMPACT	NEGATIVE IMPACT
Fluoxetine + Lamotrigine Findings: 16-week Comparative Effectiveness Trial of Lamotrigine vs. Fluoxetine for Bipolar Depression Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Fluoxetine + Aripiprazole Findings: Aripiprazole's DailyMed label states: "...lly Important Interactions with Aripiprazole Table 22: Clinically Important Drug Interactions with Aripiprazole: Concomitant Drug Name or Drug Class Clinical Rationale Clinical Recommendation Strong CYP3A4 Inhibitors (e.g., itraconazole, clarithromycin) or strong CYP2D6 inhibitors (e.g., quinidine, fluoxetine, paroxetine) Concomitant use of aripiprazole with strong CYP3A4 or CYP2D6 inhibitors increased the exposure of aripiprazole compared to the use of aripiprazole alone [see Clinical Pharmacology (12.3)]...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Fluoxetine + Oxcarbazepine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Fluoxetine + Atorvastatin Findings: Atorvastatin and Fluoxetine Prevent Oxidative Stress and Mitochondrial Dysfunction Evoked by Glutamate Toxicity in Hippocampal Slices. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
Fluoxetine + Valproic Acid Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Fluoxetine + Bupropion Findings: Bupropion's DailyMed label states: "... (7.1) • Drugs metabolized by CYP2D6: Bupropion inhibits CYP2D6 and can increase concentrations of: antidepressants (e.g., venlafaxine, nortriptyline, imipramine, desipramine, paroxetine, fluoxetine, sertraline), antipsychotics (e.g., haloperidol, risperidone, thioridazine), beta-blockers (e.g., metoprolol), and Type 1C antiarrhythmics (e.g., propafenone, flecainide). Consider dose reduction when using with bupropion...." Evidence: DailyMed label (Drug Interactions section); General web search (Tavily) Hyperlink: DailyMed
	Fluoxetine + Carbamazepine Findings: Fluoxetine's DailyMed label states: "... (7.2) Benzodiazepines: Diazepam – increased t ½, alprazolam - further psychomotor performance decrement due to increased levels. (7.7) Antipsychotics: Potential for elevation of haloperidol and

POSITIVE IMPACT	NEGATIVE IMPACT
	clozapine levels. (7.7) Anticonvulsants: Potential for elevated phenytoin and carbamazepine levels and clinical anticonvulsant toxicity. (7.7) Serotonergic Drugs: (2.9 , 2.10 , 4.1 , 5.2) Drugs that Interfere with Hemostasis (e.g., NSAIDs, Aspirin, Warfarin): May potentiate the risk of bleeding...." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Fluoxetine + Citalopram Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)
	Fluoxetine + Duloxetine Findings: Duloxetine's DailyMed label states: "...Similar effects would be expected with other potent CYP2D6 inhibitors (e.g., fluoxetine, quinidine) [see Warnings and Precautions (5.12)]" Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); General web search (Tavily) Hyperlink: DailyMed
	Fluoxetine + Escitalopram Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; External evidence (PubMed); General web search (Tavily)
	Fluoxetine + Haloperidol Findings: Fluoxetine's DailyMed label states: "... (7.2) Benzodiazepines: Diazepam – increased t ½ , alprazolam - further psychomotor performance decrement due to increased levels. (7.7) Antipsychotics: Potential for elevation of haloperidol and clozapine levels. (7.7) Anticonvulsants: Potential for elevated phenytoin and carbamazepine levels and clinical anticonvulsant toxicity...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Fluoxetine + Lithium Findings: Lithium and Fluoxetine Interaction: Risks and Management empathia.ai Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: General web search (Tavily)
	Fluoxetine + Nortriptyline Findings: Nortriptyline and Fluoxetine Interaction: Risks and Management empathia.ai Evidence: DailyMed label (Drug Interactions section); General web search (Tavily) Hyperlink: General web search (Tavily)
	Fluoxetine + Olanzapine Findings: Fluoxetine's DailyMed label states: "... (7.7) Serotonergic Drugs: (2.9 , 2.10 , 4.1 , 5.2) Drugs that Interfere with Hemostasis (e.g., NSAIDs, Aspirin, Warfarin): May potentiate the risk of bleeding. (7.4) Drugs Tightly Bound to Plasma Proteins: May cause a shift in plasma concentrations. (7.6 , 7.7) Olanzapine: When used in combination with fluoxetine, also refer to the Drug Interactions section of the package insert for Symbyax ® . (7.7) Drugs that Prolong the QT Interval: Do not use fluoxetine with thioridazine or pimozide...." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Fluoxetine + Paroxetine Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)
	Fluoxetine + Phenytoin Findings: Fluoxetine's DailyMed label states: "... (7.2) Benzodiazepines: Diazepam – increased t ½ , alprazolam - further psychomotor performance decrement due to increased levels. (7.7) Antipsychotics: Potential for elevation of haloperidol and clozapine levels. (7.7) Anticonvulsants: Potential for elevated phenytoin and carbamazepine levels and clinical anticonvulsant toxicity. (7.7) Serotonergic Drugs: (2.9 , 2.10 , 4.1 , 5.2) Drugs that Interfere with Hemostasis (e.g., NSAIDs, Aspirin, Warfarin): May potentiate the risk of bleeding...."

POSITIVE IMPACT	NEGATIVE IMPACT
	Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Fluoxetine + Quetiapine Findings: Quetiapine and Fluoxetine Interaction: Safety, Risks & Management empathia.ai Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: General web search (Tavily)
	Fluoxetine + Sertraline Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)
	Fluoxetine + Venlafaxine Findings: Venlafaxine and Fluoxetine Interaction: Risks and Management empathia.ai Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: General web search (Tavily)

HALOPERIDOL

POSITIVE IMPACT	NEGATIVE IMPACT
Haloperidol + Carbamazepine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Haloperidol + Bupropion Findings: Bupropion's DailyMed label states: "... (7.1) • Drugs metabolized by CYP2D6: Bupropion inhibits CYP2D6 and can increase concentrations of: antidepressants (e.g., venlafaxine, nortriptyline, imipramine, desipramine, paroxetine, fluoxetine, sertraline), antipsychotics (e.g., haloperidol, risperidone, thioridazine), beta-blockers (e.g., metoprolol), and Type 1C antiarrhythmics (e.g., propafenone, flecainide). Consider dose reduction when using with bupropion. (7.2) • Drugs that lower seizure threshold: Dose bupropion hydrochloride extended-release tablets (XL) with caution...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Haloperidol + Citalopram Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Haloperidol + Fluoxetine Findings: Fluoxetine's DailyMed label states: "... (7.2) Benzodiazepines: Diazepam – increased t ½ , alprazolam - further psychomotor performance decrement due to increased levels. (7.7) Antipsychotics: Potential for elevation of haloperidol and clozapine levels. (7.7) Anticonvulsants: Potential for elevated phenytoin and carbamazepine levels and clinical anticonvulsant toxicity...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Haloperidol + Duloxetine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Haloperidol + Lithium Findings: Haloperidol's DailyMed label states: "...Anticholinergic Drugs Clinical Impact The healthcare provider should keep in mind the possible increase in intraocular pressure when anticholinergic drugs are administered concomitantly with haloperidol decanoate. Prevention or Management Monitor and manage patients as clinically appropriate. Lithium Clinical Impact Concomitant use of haloperidol decanoate with lithium may cause an encephalopathic syndrome followed by irreversible brain damage [see Warnings and Precautions (5.12)]" Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Haloperidol + Escitalopram Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Haloperidol + Venlafaxine Findings: Venlafaxine's DailyMed label states: "Avoid MAOI's 14 days before starting venlafaxine and 7 days after stopping venlafaxine (5.2). Cimetidine: Caution in patients with pre-existing hypertension, in elderly patients and patients with hepatic dysfunction. (7.2) Haloperidol: Increase in Haloperidol AUC and C max . (7.4) Ketoconazole: Increase in venlafaxine and O-desmethylvenlafaxine AUC and C max . Caution when using venlafaxine with substances that inhibit both CYP2D6 and CYP3A4...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Haloperidol + Lamotrigine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	
Haloperidol + Oxcarbazepine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	
Haloperidol + Paroxetine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	
Haloperidol + Sertraline Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	

LAMOTRIGINE

POSITIVE IMPACT	NEGATIVE IMPACT
Lamotrigine + Aripiprazole Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Lamotrigine + Bupropion Findings: Massive lamotrigine and bupropion overdose resulting in status epilepticus without cardiovascular collapse. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
Lamotrigine + Duloxetine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Lamotrigine + Carbamazepine Findings: Lamotrigine's DailyMed label states: "...preparations containing 30 mcg ethinylestradiol and 150 mcg levonorgestrel ↓ levonorgestrel Decrease in levonorgestrel component by 19%. Carbamazepine and carbamazepine epoxide ↓ lamotrigine Addition of carbamazepine decreases lamotrigine concentration approximately 40%. ? carbamazepine epoxide May increase carbamazepine epoxide levels. Lopinavir/ritonavir ↓ lamotrigine Decreased lamotrigine concentration approximately 50%...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Lamotrigine + Fluoxetine Findings: 16-week Comparative Effectiveness Trial of Lamotrigine vs. Fluoxetine for Bipolar Depression Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Lamotrigine + Levetiracetam Findings: Use of Phenytoin, Phenobarbital Carbamazepine, Levetiracetam Lamotrigine and Valproate in Pregnancy and Breastfeeding: Risk of Major Malformations, Dose-dependency, Monotherapy vs Polytherapy, Pharmacokinetics and Clinical Implications. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
Lamotrigine + Haloperidol Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Lamotrigine + Phenytoin Findings: Lamotrigine's DailyMed label states: "...Lopinavir/ritonavir ↓ lamotrigine Decreased lamotrigine concentration approximately 50%. Atazanavir/ritonavir ↓ lamotrigine Decreased lamotrigine AUC approximately 32%. Phenobarbital/primidone ↓ lamotrigine Decreased lamotrigine concentration approximately 40%. Phenytoin ↓ lamotrigine Decreased lamotrigine concentration approximately 40%. Rifampin ↓ lamotrigine Decreased lamotrigine AUC approximately 40%. Valproate ↑ lamotrigine Increased lamotrigine concentrations slightly more than 2-fold...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Lamotrigine + Oxcarbazepine Findings: A Study in Participants With Epilepsy, to Evaluate the Pharmacokinetics, Safety and Tolerability of Oxcarbazepine on Padsevonil Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	
Lamotrigine + Paroxetine Findings: Efficacy and safety of two treatment algorithms in bipolar depression consisting of a combination of lithium, lamotrigine or placebo and paroxetine. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed	
Lamotrigine + Quetiapine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	
Lamotrigine + Sertraline Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	
Lamotrigine + Valproic Acid Findings: Valproic Acid's DailyMed label states: "...diazepam, ethosuximide, lamotrigine, phenytoin) by inhibiting their metabolism or protein binding displacement (7.2) Patients stabilized on rufinamide should begin valproate therapy at a low dose, and titrate to clinically effective dose (7.2) Dosage adjustment of amitriptyline/nortriptyline, propofol, warfarin, and zidovudine ma..." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); General web search (Tavily) Hyperlink: DailyMed	
Lamotrigine + Venlafaxine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	

LEVETIRACETAM

POSITIVE IMPACT	NEGATIVE IMPACT
Levetiracetam + Oxcarbazepine	Levetiracetam + Carbamazepine

POSITIVE IMPACT	NEGATIVE IMPACT
Findings: A Study in Participants With Epilepsy, to Evaluate the Pharmacokinetics, Safety and Tolerability of Oxcarbazepine on Padsevonil Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Findings: Use of Phenytoin, Phenobarbital Carbamazepine, Levetiracetam Lamotrigine and Valproate in Pregnancy and Breastfeeding: Risk of Major Malformations, Dose-dependency, Monotherapy vs Polytherapy, Pharmacokinetics and Clinical Implications. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
Levetiracetam + Quetiapine Findings: Levetiracetam adjunct to quetiapine for the acute manic phase of bipolar disorder: a randomized, double-blind and placebo-controlled clinical trial of efficacy, safety and tolerability. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed	Levetiracetam + Lamotrigine Findings: Use of Phenytoin, Phenobarbital Carbamazepine, Levetiracetam Lamotrigine and Valproate in Pregnancy and Breastfeeding: Risk of Major Malformations, Dose-dependency, Monotherapy vs Polytherapy, Pharmacokinetics and Clinical Implications. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
Levetiracetam + Valproic Acid Findings: Efficacy Comparison Between Levetiracetam and Valproic Acid in Pediatric Patients With Status Epilepticus Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Levetiracetam + Lithium Findings: Effect of U50488, a selective kappa opioid receptor agonist and levetiracetam against lithium-pilocarpine-induced status epilepticus, spontaneous convulsive seizures and related cognitive impairment. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
	Levetiracetam + Perampanel Findings: Behavioral adverse events with brivaracetam, levetiracetam, perampanel, and topiramate: A systematic review. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
	Levetiracetam + Phenytoin Findings: Use of Phenytoin, Phenobarbital Carbamazepine, Levetiracetam Lamotrigine and Valproate in Pregnancy and Breastfeeding: Risk of Major Malformations, Dose-dependency, Monotherapy vs Polytherapy, Pharmacokinetics and Clinical Implications. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
LITHIUM	
POSITIVE IMPACT	NEGATIVE IMPACT
Lithium + Aripiprazole Findings: Comparative efficacy, tolerability and acceptability of intravenous racemic ketamine with intranasal esketamine, aripiprazole and lithium as augmentative treatments for treatment-resistant unipolar depression: A systematic review and network meta-analysis. Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed	Lithium + Carbamazepine Findings: Lithium's DailyMed label states: "...Intervention: More frequent serum lithium concentration monitoring. Increase lithium dosage based on serum lithium concentration and clinical response [see Dosage and Administration (2.3)] . Methylodopa, Phenytoin and Carbamazepine Clinical Impact: Concomitant use may increase risk of adverse reactions of these drugs. Intervention: Monitor patients closely for adverse reactions of methylodopa, phenytoin, and carbamazepine. Iodide Preparations Clinical Impact: Concomitant use may produce hypothyroidism...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Lithium + Duloxetine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Lithium + Citalopram Findings: Citalopram's DailyMed label states: "...Other Serotonergic Drugs Clinical Impact: Concomitant use of citalopram and other serotonergic drugs (including other SSRIs, SNRIs, triptans, tricyclic antidepressants, opioids, lithium, buspirone, amphetamines, tryptophan, and St. John's Wort) increases the risk of serotonin syndrome. Intervention: Monitor patients for signs and symptoms of serotonin syndrome, particularly during citalopram initiation and dosage increases...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Lithium + Oxcarbazepine Findings: Comparative Efficacy and Acceptability of Antimanic Drugs in Acute Mania Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Lithium + Escitalopram Findings: Escitalopram's DailyMed label states: "...Intervention: Escitalopram tablets is contraindicated in patients taking pimozone [see Contraindications (4)]. Serotonergic Drugs Clinical Impact: Concomitant use of Escitalopram tablets and other serotonergic drugs (including other SSRIs, SNRIs, triptans, tricyclic antidepressants, opioids, lithium, buspirone, amphetamines, tryptophan, and St. John's Wort) increases the risk of serotonin syndrome. Intervention: Monitor patients for signs and symptoms of serotonin syndrome, particularly during Escitalopram tablets initiation and dosage increases...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily)
Lithium + Quetiapine Findings: Clinical and cost-effectiveness of lithium versus quetiapine augmentation for treatment-resistant depression: a pragmatic, open-label, parallel-group, randomised controlled superiority trial in the UK. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed	
Lithium + Valproic Acid Findings: Cost- Effectiveness and Quality of Life Assessment in Mood Disorder Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	

POSITIVE IMPACT	NEGATIVE IMPACT
	Hyperlink: DailyMed
	Lithium + Fluoxetine Findings: Lithium and Fluoxetine Interaction: Risks and Management empathia.ai Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: General web search (Tavily)
	Lithium + Haloperidol Findings: Haloperidol's DailyMed label states: "...Anticholinergic Drugs Clinical Impact The healthcare provider should keep in mind the possible increase in intraocular pressure when anticholinergic drugs are administered concomitantly with haloperidol decanoate. Prevention or Management Monitor and manage patients as clinically appropriate. Lithium Clinical Impact Concomitant use of haloperidol decanoate with lithium may cause an encephalopathic syndrome followed by irreversible brain damage [see Warnings and Precautions (5.12)]" Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Lithium + Levetiracetam Findings: Effect of U50488, a selective kappa opioid receptor agonist and levetiracetam against lithium-pilocarpine-induced status epilepticus, spontaneous convulsive seizures and related cognitive impairment. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
	Lithium + Paroxetine Findings: Paroxetine's DailyMed label states: "...If serotonin syndrome occurs, consider discontinuation of paroxetine and/or concomitant serotonergic drugs [see Warnings and Precautions (5.2)] . Examples other SSRIs, SNRIs, triptans, tricyclic antidepressants, opioids, lithium, tryptophan, buspirone, amphetamines, and St. John's Wort. Drugs that Interfere with Hemostasis (antiplatelet agents and anticoagulants) Clinical Impact The concurrent use of an antiplatelet agent or anticoagulant with paroxetine may potentiate the risk of bleeding...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Lithium + Phenytoin Findings: Lithium's DailyMed label states: "...Intervention: More frequent serum lithium concentration monitoring. Increase lithium dosage based on serum lithium concentration and clinical response [see Dosage and Administration (2.3)] . Methyldopa, Phenytoin and Carbamazepine Clinical Impact: Concomitant use may increase risk of adverse reactions of these drugs. Intervention: Monitor patients closely for adverse reactions of methyldopa, phenytoin, and carbamazepine. Iodide Preparations Clinical Impact: Concomitant use may produce hypothyroidism...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Lithium + Sertraline Findings: Sertraline's DailyMed label states: "...If serotonin syndrome occurs, consider discontinuation of sertraline hydrochloride and/or concomitant serotonergic drugs [See Warnings and Precautions (5.2)] . Examples: Other SSRIs, SNRIs, triptans, tricyclic antidepressants, opioids, lithium, tryptophan, buspirone, amphetamines, and St. John's Wort. Drugs that Interfere with Hemostasis (antiplatelet agents and anticoagulants) Clinical Impact: The concurrent use of an antiplatelet agent or anticoagulant with sertraline hydrochloride may potentiate the risk of bleeding...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Lithium + Venlafaxine Findings: Venlafaxine's DailyMed label states: "...Caution should be exercised with co-administration of venlafaxine and metoprolol. (7.8) CNS-active drugs: Caution when using venlafaxine with such drugs. (7.10) Serotonergic drugs (e.g., triptans, SSRIs, other SNRIs, linezolid, lithium, tramadol, or St. John's Wort): Potential for serotonin syndrome. Careful patient observation advised. (7.10) Tryptophan supplements: Concomitant use not recommended...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed

METHYLPHENIDATE

POSITIVE IMPACT	NEGATIVE IMPACT
Methylphenidate + Valproic Acid Findings: Improvement by methylphenidate and atomoxetine of social interaction deficits and recognition memory impairment in a mouse model of valproic acid-induced autism. Evidence: External evidence (PubMed); General web search (Tavily) Hyperlink: PubMed	Methylphenidate + Atomoxetine Findings: Methylphenidate and Atomoxetine in Pregnancy and Possible Adverse Fetal Outcomes: A Systematic Review and Meta-Analysis. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
	Methylphenidate + Phenytoin Findings: Phenytoin's DailyMed label states: "...oxamine, sertraline Gastric acid reducing agents H 2 antagonists (cimetidine), omeprazole Sulfonamides Sulfamethizole, sulfaphenazole, sulfadiazine, sulfamethoxazole-trimethoprim Other Acute alcohol intake, amiodarone, chloramphenicol, chlordiazepoxide, disulfiram, estrogen, fluvastatin, isoniazid, methylphenidate, phenothiazines, salicylates, ticlopidine, tolbutamide, trazodone, warfarin Drugs that may decrease phenytoin serum levels Antacids Antacids may affect absorption of phenytoin...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); General web search (Tavily) Hyperlink: DailyMed

NORTRIPTYLINE

POSITIVE IMPACT	NEGATIVE IMPACT
Nortriptyline + Escitalopram Findings: Effectiveness of Escitalopram and Nortriptyline on Depressive Symptoms in Parkinson's disease: the ADepT-PD RCT pilot. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed	Nortriptyline + Bupropion Findings: Bupropion's DailyMed label states: "... (7.1) • Drugs metabolized by CYP2D6: Bupropion inhibits CYP2D6 and can increase concentrations of: antidepressants (e.g., venlafaxine, nortriptyline, imipramine, desipramine, paroxetine, fluoxetine, sertraline), antipsychotics (e.g., haloperidol, risperidone, thioridazine), beta-blockers (e.g., metoprolol), and Type 1C antiarrhythmics (e.g., propafenone, flecainide). Consider dose reduction when using with bupropion...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Nortriptyline + Sertraline Findings: Cost- Effectiveness and Quality of Life Assessment in Major Depression Disorder Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Nortriptyline + Fluoxetine Findings: Nortriptyline and Fluoxetine Interaction: Risks and Management empathia.ai Evidence: DailyMed label (Drug Interactions section); General web search (Tavily) Hyperlink: General web search (Tavily)
Nortriptyline + Valproic Acid Findings: Valproic Acid's DailyMed label states: "...diazepam, ethosuximide, lamotrigine, phenytoin) by inhibiting their metabolism or protein binding displacement (7.2) Patients stabilized on rufinamide should begin valproate therapy at a low dose, and titrate to clinically effective dose (7.2) Dosage adjustment of amitriptyline/nortriptyline, propofol, warfarin, and zidovudine may be necessary if used concomitantly with valproic acid (7.2) Topiramate: Hyperammonemia and encephalopathy (5.10 , 7.3) 7.1 Effects of Co-Administered Drugs on Valproate Clearance Drugs that affect the level of expression of hepatic enzymes, particularly t..." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed	Nortriptyline + Paroxetine Findings: Nortriptyline and Paroxetine Interaction: Risks and Management empathia.ai Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: General web search (Tavily)
Nortriptyline + Venlafaxine Findings: Efficacy and safety of venlafaxine versus nortriptyline for the preventive treatment of migraine: A double-blind randomized clinical trial. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed	

OLANZAPINE

POSITIVE IMPACT	NEGATIVE IMPACT
Olanzapine + Bupropion Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Olanzapine + Aripiprazole Findings: Pregnancy exposure to olanzapine, quetiapine, risperidone, aripiprazole and risk of congenital malformations. A systematic review. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
Olanzapine + Duloxetine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Olanzapine + Carbamazepine Findings: Olanzapine's DailyMed label states: "Diazepam: May potentiate orthostatic hypotension (7.1, 7.2) Alcohol: May potentiate orthostatic hypotension (7.1) Carbamazepine: Increased clearance of olanzapine (7.1) Fluvoxamine: May increase olanzapine levels (7.1) Olanzapine and Fluoxetine in Combination: Also refer to the DrugInteractions section of the package insert for Symbyax...."
Olanzapine + Escitalopram Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily)	

POSITIVE IMPACT	NEGATIVE IMPACT
Hyperlink: ClinicalTrials.gov Olanzapine + Lamotrigine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov Olanzapine + Oxcarbazepine Findings: Comparative Efficacy and Acceptability of Antimanic Drugs in Acute Mania Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed Olanzapine + Fluoxetine Findings: Fluoxetine's DailyMed label states: "... (7.7) Serotonergic Drugs: (2.9 , 2.10 , 4.1 , 5.2) Drugs that Interfere with Hemostasis (e.g., NSAIDs, Aspirin, Warfarin): May potentiate the risk of bleeding. (7.4) Drugs Tightly Bound to Plasma Proteins: May cause a shift in plasma concentrations. (7.6 , 7.7) Olanzapine: When used in combination with fluoxetine, also refer to the Drug Interactions section of the package insert for Symbyax ® . (7.7) Drugs that Prolong the QT Interval: Do not use fluoxetine with thioridazine or pimozide...." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed

OXCARBAZEPINE

POSITIVE IMPACT	NEGATIVE IMPACT
Oxcarbazepine + Aripiprazole Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov Oxcarbazepine + Bupropion Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov Oxcarbazepine + Carbamazepine Findings: Oxcarbazepine's DailyMed label states: "(2.5 , 7.2) Oral Contraceptives : Advise patients that Oxtellar XR may decrease the effectiveness of hormonal contraceptives...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); General web search (Tavily) Hyperlink: DailyMed Oxcarbazepine + Citalopram Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov Oxcarbazepine + Duloxetine Findings: Comparative efficacy and safety of gabapentin, pregabalin, oxcarbazepine, and duloxetine in diabetic peripheral neuropathy: A network meta-analysis. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed Oxcarbazepine + Escitalopram Findings: Comparison of clinical efficacy of oxcarbazepine and lamotrigine combined with escitalopram, and impact on prognostic quality of life in treating patients with epilepsy and depressive disorder. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed Oxcarbazepine + Fluoxetine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov Oxcarbazepine + Haloperidol Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov Oxcarbazepine + Lamotrigine Findings: A Study in Participants With Epilepsy, to Evaluate the Pharmacokinetics, Safety and Tolerability of Oxcarbazepine on Padsevonil Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov Oxcarbazepine + Levetiracetam	Oxcarbazepine + Atomoxetine Findings: Acute oxcarbazepine and atomoxetine overdose with quetiapine. Evidence: External evidence (PubMed); General web search (Tavily) Hyperlink: PubMed Oxcarbazepine + Quetiapine Findings: Acute oxcarbazepine and atomoxetine overdose with quetiapine. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed

POSITIVE IMPACT	NEGATIVE IMPACT
Findings: A Study in Participants With Epilepsy, to Evaluate the Pharmacokinetics, Safety and Tolerability of Oxcarbazepine on Padsevonil Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	
Oxcarbazepine + Lithium Findings: Comparative Efficacy and Acceptability of Antimanic Drugs in Acute Mania Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	
Oxcarbazepine + Olanzapine Findings: Comparative Efficacy and Acceptability of Antimanic Drugs in Acute Mania Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	
Oxcarbazepine + Perampanel Findings: Perampanel's DailyMed label states: "7 DRUG INTERACTIONS Contraceptives: 12 mg once daily may decrease the effectiveness of hormonal contraceptives containing levonorgestrel (7.1) Moderate and Strong CYP3A4 Inducers (including carbamazepine, oxcarbazepine, and phenytoin): increase clearance of perampanel and decrease perampanel plasma concentrations. When moderate or strong CYP3A4 inducers are introduced or withdrawn, monitor patients closely...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed	
Oxcarbazepine + Phenytoin Findings: Oxcarbazepine's DailyMed label states: "(2.5 , 7.2) Oral Contraceptives : Advise patients that Oxtellar XR may decrease the effectiveness of hormonal contraceptives...." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed	

PAROXETINE

POSITIVE IMPACT	NEGATIVE IMPACT
Paroxetine + Carbamazepine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Paroxetine + Aripiprazole Findings: Aripiprazole's DailyMed label states: "...t Interactions with Aripiprazole Table 22: Clinically Important Drug Interactions with Aripiprazole: Concomitant Drug Name or Drug Class Clinical Rationale Clinical Recommendation Strong CYP3A4 Inhibitors (e.g., itraconazole, clarithromycin) or strong CYP2D6 inhibitors (e.g., quinidine, fluoxetine, paroxetine) Concomitant use of aripiprazole with strong CYP3A4 or CYP2D6 inhibitors increased the exposure of aripiprazole compared to the use of aripiprazole alone [see Clinical Pharmacology (12.3)]...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Paroxetine + Haloperidol Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Paroxetine + Atomoxetine Findings: Paroxetine's DailyMed label states: "...Intervention Decrease the dosage of a CYP2D6 substrate if needed with concomitant paroxetine use. Conversely, an increase in dosage of a CYP2D6 substrate may be needed if paroxetine is discontinued. Examples propafenone, flecainide, atomoxetine, desipramine, dextromethorphan, metoprolol, nebivolol, perphenazine, tolterodine, venlafaxine, risperidone...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); General web search (Tavily) Hyperlink: DailyMed
Paroxetine + Lamotrigine Findings: Efficacy and safety of two treatment algorithms in bipolar depression consisting of a combination of lithium, lamotrigine or placebo and paroxetine. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed	Paroxetine + Bupropion Findings: Bupropion's DailyMed label states: "... (7.1) • Drugs metabolized by CYP2D6: Bupropion inhibits CYP2D6 and can increase concentrations of: antidepressants (e.g., venlafaxine, nortriptyline, imipramine, desipramine, paroxetine, fluoxetine, sertraline), antipsychotics (e.g., haloperidol, risperidone, thioridazine), beta-blockers (e.g., metoprolol), and Type 1C antiarrhythmics (e.g., propafenone, flecainide). Consider dose reduction when using with bupropion...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Paroxetine + Citalopram

POSITIVE IMPACT	NEGATIVE IMPACT
	Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; ClinicalTrials.gov; General web search (Tavily)
	Paroxetine + Duloxetine Findings: Duloxetine's DailyMed label states: "...Other drugs that inhibit CYP1A2 metabolism include cimetidine and quinolone antimicrobials such as ciprofloxacin and enoxacin [see Warnings and Precautions (5.12)] . 7.2 Inhibitors of CYP2D6 Concomitant use of duloxetine (40 mg once daily) with paroxetine (20 mg once daily) increased the concentration of duloxetine AUC by about 60%, and greater degrees of inhibition are expected with higher doses of paroxetine...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Paroxetine + Escitalopram Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)
	Paroxetine + Fluoxetine Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)
	Paroxetine + Lithium Findings: Paroxetine's DailyMed label states: "...If serotonin syndrome occurs, consider discontinuation of paroxetine and/or concomitant serotonergic drugs [see Warnings and Precautions (5.2)] . Examples other SSRIs, SNRIs, triptans, tricyclic antidepressants, opioids, lithium, tryptophan, buspirone, amphetamines, and St. John's Wort. Drugs that Interfere with Hemostasis (antiplatelet agents and anticoagulants) Clinical Impact The concurrent use of an antiplatelet agent or anticoagulant with paroxetine may potentiate the risk of bleeding...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Paroxetine + Nortriptyline Findings: Nortriptyline and Paroxetine Interaction: Risks and Management empathia.ai Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: General web search (Tavily)
	Paroxetine + Sertraline Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)
	Paroxetine + Venlafaxine Findings: Paroxetine's DailyMed label states: "...Conversely, an increase in dosage of a CYP2D6 substrate may be needed if paroxetine is discontinued. Examples propafenone, flecainide, atomoxetine, desipramine, dextromethorphan, metoprolol, nebivolol, perphenazine, tolterodine, venlafaxine, risperidone...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed

PERAMPANEL

POSITIVE IMPACT	NEGATIVE IMPACT
Perampanel + Carbamazepine Findings: Perampanel's DailyMed label states: "7 DRUG INTERACTIONS Contraceptives: 12 mg once daily may decrease the effectiveness of hormonal contraceptives containing levonorgestrel (7.1) Moderate and Strong CYP3A4 Inducers (including carbamazepine, oxcarbazepine, and phenytoin): increase clearance of perampanel and decrease perampanel plasma concentrations. When moderate or strong CYP3A4 inducers are introduced or withdrawn, monitor patients closely...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed	Perampanel + Levetiracetam Findings: Behavioral adverse events with brivaracetam, levetiracetam, perampanel, and topiramate: A systematic review. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed

POSITIVE IMPACT	NEGATIVE IMPACT
Perampanel + Oxcarbazepine Findings: Perampanel's DailyMed label states: "7 DRUG INTERACTIONS Contraceptives: 12 mg once daily may decrease the effectiveness of hormonal contraceptives containing levonorgestrel (7.1) Moderate and Strong CYP3A4 Inducers (including carbamazepine, oxcarbazepine, and phenytoin): increase clearance of perampanel and decrease perampanel plasma concentrations. When moderate or strong CYP3A4 inducers are introduced or withdrawn, monitor patients closely...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed	
Perampanel + Phenytoin Findings: Perampanel's DailyMed label states: "7 DRUG INTERACTIONS Contraceptives: 12 mg once daily may decrease the effectiveness of hormonal contraceptives containing levonorgestrel (7.1) Moderate and Strong CYP3A4 Inducers (including carbamazepine, oxcarbazepine, and phenytoin): increase clearance of perampanel and decrease perampanel plasma concentrations. When moderate or strong CYP3A4 inducers are introduced or withdrawn, monitor patients closely...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed	

PHENYTOIN

POSITIVE IMPACT	NEGATIVE IMPACT
Phenytoin + Oxcarbazepine Findings: Oxcarbazepine's DailyMed label states: "(2.5 , 7.2) Oral Contraceptives : Advise patients that Oxtellar XR may decrease the effectiveness of hormonal contraceptives...." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed	Phenytoin + Bupropion Findings: Bupropion's DailyMed label states: "(7.1) • Drugs metabolized by CYP2D6: Bupropion inhibits CYP2D6 and can increase concentrations of: antidepressants (e.g., venlafaxine, nortriptyline, imipramine, desipramine, paroxetine, fluoxetine, sertraline), anti..." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Phenytoin + Perampanel Findings: Perampanel's DailyMed label states: "7 DRUG INTERACTIONS Contraceptives: 12 mg once daily may decrease the effectiveness of hormonal contraceptives containing levonorgestrel (7.1) Moderate and Strong CYP3A4 Inducers (including carbamazepine, oxcarbazepine, and phenytoin): increase clearance of perampanel and decrease perampanel plasma concentrations. When moderate or strong CYP3A4 inducers are introduced or withdrawn, monitor patients closely...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed	Phenytoin + Carbamazepine Findings: Carbamazepine's DailyMed label states: "...Agents that are CYP inducers that have been found, or are expected, to decrease plasma levels of carbamazepine extended-release capsules include, for example, the following: Aminophylline, cisplatin, doxorubicin HCL, felbamate, fosphenytoin, methsuximide, phenobarbital, phenytoin(1), primidone, rifampin and theophylline. (1) Phenytoin plasma levels have also been reported to increase and decrease in the presence of carbamazepine, see below...." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Phenytoin + Ethosuximide Findings: Ethosuximide's DailyMed label states: "Drug Interactions Since ethosuximide may interact with concurrently administered antiepileptic drugs, periodic serum level determinations of these drugs may be necessary (e.g., ethosuximide may elevate phenytoin serum levels and valproic acid has been reported to both increase and decrease ethosuximide levels)." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Phenytoin + Fluoxetine Findings: Fluoxetine's DailyMed label states: "... (7.2) Benzodiazepines: Diazepam – increased t ½ , alprazolam - further psychomotor performance decrement due to increased levels. (7.7) Antipsychotics: Potential for elevation of haloperidol and clozapine levels. (7.7) Anticonvulsants: Potential for elevated phenytoin and carbamazepine levels and clinical anticonvulsant toxicity. (7.7) Serotonergic Drugs: (2.9 , 2.10 , 4.1 , 5.2) Drugs that Interfere with Hemostasis (e.g., NSAIDs, Aspirin, Warfarin): May potentiate the risk of bleeding...." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed

POSITIVE IMPACT	NEGATIVE IMPACT
	Phenytoin + Lamotrigine Findings: Lamotrigine's DailyMed label states: "...Lopinavir/ritonavir ↓ lamotrigine Decreased lamotrigine concentration approximately 50%. Atazanavir/ritonavir ↓ lamotrigine Decreased lamotrigine AUC approximately 32%. Phenobarbital/primidone ↓ lamotrigine Decreased lamotrigine concentration approximately 40%. Phenytoin ↓ lamotrigine Decreased lamotrigine concentration approximately 40%. Rifampin ↓ lamotrigine Decreased lamotrigine AUC approximately 40%. Valproate ↑ lamotrigine Increased lamotrigine concentrations slightly more than 2-fold...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Phenytoin + Levetiracetam Findings: Use of Phenytoin, Phenobarbital Carbamazepine, Levetiracetam Lamotrigine and Valproate in Pregnancy and Breastfeeding: Risk of Major Malformations, Dose-dependency, Monotherapy vs Polytherapy, Pharmacokinetics and Clinical Implications. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
	Phenytoin + Lithium Findings: Lithium's DailyMed label states: "...Intervention: More frequent serum lithium concentration monitoring. Increase lithium dosage based on serum lithium concentration and clinical response [see Dosage and Administration (2.3)] . Methylodopa, Phenytoin and Carbamazepine Clinical Impact: Concomitant use may increase risk of adverse reactions of these drugs. Intervention: Monitor patients closely for adverse reactions of methylodopa, phenytoin, and carbamazepine. Iodide Preparations Clinical Impact: Concomitant use may produce hypothyroidism...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Phenytoin + Methylphenidate Findings: Phenytoin's DailyMed label states: "...oxamine, sertraline Gastric acid reducing agents H 2 antagonists (cimetidine), omeprazole Sulfonamides Sulfamethizole, sulfaphenazole, sulfadiazine, sulfamethoxazole-trimethoprim Other Acute alcohol intake, amiodarone, chloramphenicol, chlordiazepoxide, disulfiram, estrogen, fluvastatin, isoniazid, methylphenidate, phenothiazines, salicylates, ticlopidine, tolbutamide, trazodone, warfarin Drugs that may decrease phenytoin serum levels Antacids Antacids may affect absorption of phenytoin...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); General web search (Tavily) Hyperlink: DailyMed
	Phenytoin + Quetiapine Findings: Quetiapine's DailyMed label states: "...one sixth when coadministered with strong CYP3A4 inhibitors (e.g., ketoconazole, ritonavir) (2.5 , 7.1 , 12.3) Concomitant use of strong CYP3A4 inducers: Increase quetiapine dose up to 5 fold when used in combination with a chronic treatment (more than 7-14 days) of potent CYP3A4 inducers (e.g., phenytoin, rifampin, St...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Phenytoin + Sertraline Findings: Phenytoin's DailyMed label states: "...ples Drugs that may increase phenytoin serum levels Antiepileptic drugs Ethosuximide, felbamate, oxcarbazepine, methsuximide, topiramate Azoles Fluconazole, ketoconazole, itraconazole, miconazole, voriconazole Antineoplastic agents Capecitabine, fluorouracil Antidepressants Fluoxetine, fluvoxamine, sertraline Gastric acid reducing agents H 2 antagonists (cimetidine), omeprazole Sulfonamides Sulfamethizole, sulfaphenazole, sulfadiazine, sulfamethoxazole-trimethoprim Other Acute alcohol intake, amiodarone, chloramphenicol, chlordiazepoxide, disulfiram, estrogen, fluvastatin, isoniazid, methylphenidate, ph..." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Phenytoin + Valproic Acid Findings: Phenytoin's DailyMed label states: "...John's wort The induction potency of St. John's wort may vary widely based on preparation. , sucralfate, theophylline Drugs that may either increase or decrease phenytoin serum levels Antiepileptic drugs

POSITIVE IMPACT	NEGATIVE IMPACT
	Phenobarbital, valproate sodium Valproate sodium and valproic acid are similar medications. The term valproate has been used to represent these medications. , valproic acid 7.2 Drugs Affected by Phenytoin Table 3 includes commonly occurring drug interactions affected by phenytoin. However, this list is not intended to be inclusive or comprehensive...." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); General web search (Tavily) Hyperlink: DailyMed

QUETIAPINE

POSITIVE IMPACT	NEGATIVE IMPACT
Quetiapine + Duloxetine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Quetiapine + Aripiprazole Findings: Pregnancy exposure to olanzapine, quetiapine, risperidone, aripiprazole and risk of congenital malformations. A systematic review. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
Quetiapine + Lamotrigine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Quetiapine + Atomoxetine Findings: Acute oxcarbazepine and atomoxetine overdose with quetiapine. Evidence: External evidence (PubMed); General web search (Tavily) Hyperlink: PubMed
Quetiapine + Levetiracetam Findings: Levetiracetam adjunct to quetiapine for the acute manic phase of bipolar disorder: a randomized, double-blind and placebo-controlled clinical trial of efficacy, safety and tolerability. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed	Quetiapine + Atorvastatin Findings: Adverse effects of interactions between antipsychotics and ... Evidence: General web search (Tavily) Hyperlink: General web search (Tavily)
Quetiapine + Lithium Findings: Clinical and cost-effectiveness of lithium versus quetiapine augmentation for treatment-resistant depression: a pragmatic, open-label, parallel-group, randomised controlled superiority trial in the UK. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed	Quetiapine + Carbamazepine Findings: Quetiapine's DailyMed label states: "...Quetiapine exposure is increased by the prototype CYP3A4 inhibitors (e.g., ketoconazole, itraconazole, indinavir, ritonavir, nefazodone, etc.) and decreased by the prototype CYP3A4 inducers (e.g., phenytoin, carbamazepine, rifampin, avasimibe, St. John's wort etc.). Dose adjustment of quetiapine will be necessary if it is co-administered with potent CYP3A4 inducers or inhibitors...." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Quetiapine + Clobazam Findings: Central Pontine Myelinolysis: A Rare and Life-Threatening Adverse Effect of Clobazam and Quetiapine. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
	Quetiapine + Fluoxetine Findings: Quetiapine and Fluoxetine Interaction: Safety, Risks & Management empathia.ai Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: General web search (Tavily)
	Quetiapine + Oxcarbazepine Findings: Acute oxcarbazepine and atomoxetine overdose with quetiapine. Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed
	Quetiapine + Phenytoin Findings: Quetiapine's DailyMed label states: "...one sixth when coadministered with strong CYP3A4 inhibitors (e.g., ketoconazole, ritonavir) (2.5 , 7.1 , 12.3) Concomitant use of strong CYP3A4 inducers: Increase quetiapine dose up to 5 fold when used in combination with a chronic treatment (more than 7-14 days) of potent CYP3A4 inducers (e.g., phenytoin, rifampin, St...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Quetiapine + Sertraline Findings: Quetiapine and Sertraline: Interaction Risks & Advice Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: General web search (Tavily)

POSITIVE IMPACT	NEGATIVE IMPACT
	Quetiapine + Venlafaxine Findings: Quetiapine and Venlafaxine: Drug Interaction Risks Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: General web search (Tavily)

SERTRALINE

POSITIVE IMPACT	NEGATIVE IMPACT
Sertraline + Aripiprazole Findings: Aripiprazole/Sertraline Combination: Clinical and Cost-Effectiveness in Comparison With Quetiapine for the Treatment of Bipolar Depression (ASCeND Trial)-Protocol for a Nested Qualitative Study. Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: PubMed	Sertraline + Atomoxetine Findings: Sertraline's DailyMed label states: "...Intervention: Decrease the dosage of a CYP2D6 substrate if needed with concomitant sertraline hydrochloride use. Conversely, an increase in dosage of a CYP2D6 substrate may be needed if sertraline hydrochloride is discontinued. Examples: propafenone, flecainide, atomoxetine, desipramine, dextromethorphan, metoprolol, nebivolol, perphenazine, thioridazine, tolterodine, venlafaxine Phenytoin Clinical Impact: Phenytoin is a narrow therapeutic index drug. Sertraline hydrochloride may increase phenytoin concentrations...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Sertraline + Carbamazepine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Sertraline + Bupropion Findings: Bupropion's DailyMed label states: "... (7.1) • Drugs metabolized by CYP2D6: Bupropion inhibits CYP2D6 and can increase concentrations of: antidepressants (e.g., venlafaxine, nortriptyline, imipramine, desipramine, paroxetine, fluoxetine, sertraline), antipsychotics (e.g., haloperidol, risperidone, thioridazine), beta-blockers (e.g., metoprolol), and Type 1C antiarrhythmics (e.g., propafenone, flecainide). Consider dose reduction when using with bupropion...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Sertraline + Haloperidol Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Sertraline + Citalopram Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)
Sertraline + Lamotrigine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Sertraline + Duloxetine Findings: Duloxetine and Sertraline Interaction: Risks and Management empathia.ai Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: General web search (Tavily)
Sertraline + Nortriptyline Findings: Cost- Effectiveness and Quality of Life Assessment in Major Depression Disorder Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Sertraline + Escitalopram Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)
	Sertraline + Fluoxetine Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)
	Sertraline + Lithium Findings: Sertraline's DailyMed label states: "...If serotonin syndrome occurs, consider discontinuation of sertraline hydrochloride and/or concomitant serotonergic drugs [See Warnings and Precautions (5.2)] . Examples: Other SSRIs, SNRIs, triptans, tricyclic antidepressants, opioids, lithium, tryptophan, buspirone, amphetamines, and St. John's Wort. Drugs that Interfere with Hemostasis (antiplatelet agents and anticoagulants) Clinical Impact: The concurrent use of an antiplatelet agent or anticoagulant with sertraline hydrochloride may potentiate the risk of bleeding...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Sertraline + Paroxetine Findings: Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. Evidence: Pharmacodynamic interaction reference; External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)
	Sertraline + Phenytoin Findings: Phenytoin's DailyMed label states: "...ples Drugs that may increase phenytoin serum levels Antiepileptic drugs Ethosuximide, felbamate, oxcarbazepine, methsuximide, topiramate Azoles Fluconazole, ketoconazole, itraconazole,

POSITIVE IMPACT	NEGATIVE IMPACT
	miconazole, voriconazole Antineoplastic agents Capecitabine, fluorouracil Antidepressants Fluoxetine, fluvoxamine, sertraline Gastric acid reducing agents H 2 antagonists (cimetidine), omeprazole Sulfonamides Sulfamethizole, sulfaphenazole, sulfadiazine, sulfamethoxazole-trimethoprim Other Acute alcohol intake, amiodarone, chloramphenicol, chlordiazepoxide, disulfiram, estrogen, fluvastatin, isoniazid, methylphenidate, ph..." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Sertraline + Quetiapine Findings: Quetiapine and Sertraline: Interaction Risks & Advice Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: General web search (Tavily)
	Sertraline + Venlafaxine Findings: Sertraline's DailyMed label states: "...Conversely, an increase in dosage of a CYP2D6 substrate may be needed if sertraline hydrochloride is discontinued. Examples: propafenone, flecainide, atomoxetine, desipramine, dextromethorphan, metoprolol, nebivolol, perphenazine, thioridazine, tolterodine, venlafaxine Phenytoin Clinical Impact: Phenytoin is a narrow therapeutic index drug. Sertraline hydrochloride may increase phenytoin concentrations. Intervention: Monitor phenytoin levels when initiating or titrating sertraline hydrochloride. Reduce phenytoin dosage if needed...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed

VALPROIC ACID

POSITIVE IMPACT	NEGATIVE IMPACT
Valproic Acid + Atomoxetine Findings: Improvement by methylphenidate and atomoxetine of social interaction deficits and recognition memory impairment in a mouse model of valproic acid-induced autism. Evidence: External evidence (PubMed); General web search (Tavily) Hyperlink: PubMed	Valproic Acid + Carbamazepine Findings: Carbamazepine's DailyMed label states: "...Accordingly, the dosage of carbamazepine extended-release capsules should be adjusted and/or the plasma levels monitored when used concomitantly with loxapine, quetiapine, or valproic acid. Thus, if a patient has been titrated to a stable dosage of carbamazepine extended-release capsules, and then begins a course of treatment with one of these CYP3A4 or epoxide hydrolase inhibitors, it is reasonable to expect that a dose reduction for carbamazepine extended-release capsules may be ne..." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Valproic Acid + Bupropion Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Valproic Acid + Ethosuximide Findings: Ethosuximide's DailyMed label states: "Drug Interactions Since ethosuximide may interact with concurrently administered antiepileptic drugs, periodic serum level determinations of these drugs may be necessary (e.g., ethosuximide may elevate phenytoin serum levels and valproic acid has been reported to both increase and decrease ethosuximide levels)." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
Valproic Acid + Citalopram Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	Valproic Acid + Phenytoin Findings: Phenytoin's DailyMed label states: "...John's wort The induction potency of St. John's wort may vary widely based on preparation. , sucralfate, theophylline Drugs that may either increase or decrease phenytoin serum levels Antiepileptic drugs Phenobarbital, valproate sodium Valproate sodium and valproic acid are similar medications. The term valproate has been used to represent these medications. , valproic acid 7.2 Drugs Affected by Phenytoin Table 3 includes commonly occurring drug interactions affected by phenytoin. However, this list is not intended to be inclusive or comprehensive...." Evidence: DailyMed label (Drug Interactions section); DailyMed label (Drug Interactions section); External evidence (PubMed); General web search (Tavily) Hyperlink: DailyMed
Valproic Acid + Duloxetine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	
Valproic Acid + Escitalopram Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	
Valproic Acid + Fluoxetine Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: ClinicalTrials.gov	
Valproic Acid + Lamotrigine Findings: Valproic Acid's DailyMed label states: "...diazepam, ethosuximide, lamotrigine, phenytoin) by inhibiting their metabolism or protein binding displacement (7.2) Patients stabilized on rufinamide should begin valproate therapy at a low dose, and titrate to clinically effective dose (7.2) Dosage adjustment of amitriptyline/nortriptyline, propofol, warfarin, and zidovudine ma..." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); General web search (Tavily)	

POSITIVE IMPACT	NEGATIVE IMPACT
Hyperlink: DailyMed	
Valproic Acid + Levetiracetam	
Findings: Efficacy Comparison Between Levetiracetam and Valproic Acid in Pediatric Patients With Status Epilepticus	
Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)	
Hyperlink: ClinicalTrials.gov	
Valproic Acid + Lithium	
Findings: Cost- Effectiveness and Quality of Life Assessment in Mood Disorder	
Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)	
Hyperlink: ClinicalTrials.gov	
Valproic Acid + Methylphenidate	
Findings: Improvement by methylphenidate and atomoxetine of social interaction deficits and recognition memory impairment in a mouse model of valproic acid-induced autism.	
Evidence: External evidence (PubMed); General web search (Tavily)	
Hyperlink: PubMed	
Valproic Acid + Nortriptyline	
Findings: Valproic Acid's DailyMed label states: "...diazepam, ethosuximide, lamotrigine, phenytoin) by inhibiting their metabolism or protein binding displacement (7.2) Patients stabilized on rufinamide should begin valproate therapy at a low dose, and titrate to clinically effective dose (7.2) Dosage adjustment of amitriptyline/nortriptyline, propofol, warfarin, and zidovudine may be necessary if used concomitantly with valproic acid (7.2) Topiramate: Hyperammonemia and encephalopathy (5.10 , 7.3) 7.1 Effects of Co-Administered Drugs on Valproate Clearance Drugs that affect the level of expression of hepatic enzymes, particularly t..."	
Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily)	
Hyperlink: DailyMed	
Valproic Acid + Venlafaxine	
Findings: The Efficacy of Venlafaxine, Flunarizine, and Valproic Acid in the Prophylaxis of Vestibular Migraine.	
Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)	
Hyperlink: PubMed	

VENLAFAXINE

POSITIVE IMPACT	NEGATIVE IMPACT
Venlafaxine + Aripiprazole	
Findings: Comparative Effectiveness Research Trial for Antidepressant Incomplete and Nonresponders With Treatment Resistant Depression (ASCERTAIN-TRD): Effect of Aripiprazole or Repetitive Transcranial Magnetic Stimulation Augmentation Versus Switching to the Antidepressant Venlafaxine on Quality of Life.	
Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)	
Hyperlink: PubMed	
Venlafaxine + Carbamazepine	
Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies	
Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)	
Hyperlink: ClinicalTrials.gov	
Venlafaxine + Lamotrigine	
Findings: Longitudinal Comparative Effectiveness of Bipolar Disorder Therapies	
Evidence: ClinicalTrials.gov; General web search (Tavily)	
Hyperlink: ClinicalTrials.gov	
Venlafaxine + Nortriptyline	
Findings: Efficacy and safety of venlafaxine versus nortriptyline for the preventive treatment of migraine: A double-blind randomized clinical trial.	
Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)	
Hyperlink: PubMed	
Venlafaxine + Valproic Acid	
Findings: The Efficacy of Venlafaxine, Flunarizine, and Valproic Acid in the Prophylaxis of Vestibular Migraine.	
Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)	
Venlafaxine + Atorvastatin	
Findings: Atorvastatin Interactions: Risks, Effects, & Recommendations	
Evidence: General web search (Tavily)	
Hyperlink: General web search (Tavily)	
Venlafaxine + Bupropion	
Findings: Bupropion's DailyMed label states: "... (7.1) • Drugs metabolized by CYP2D6: Bupropion inhibits CYP2D6 and can increase concentrations of: antidepressants (e.g., venlafaxine, nortriptyline, imipramine, desipramine, paroxetine, fluoxetine, sertraline), antipsychotics (e.g., haloperidol, risperidone, thioridazine), beta-blockers (e.g., metoprolol), and Type 1C antiarrhythmics (e.g., propafenone, flecainide). Consider dose reduction when using with bupropion...."	
Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily)	
Hyperlink: DailyMed	
Venlafaxine + Fluoxetine	
Findings: Venlafaxine and Fluoxetine Interaction: Risks and Management empathia.ai	
Evidence: External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily)	
Hyperlink: General web search (Tavily)	
Venlafaxine + Haloperidol	
Findings: Venlafaxine's DailyMed label states: "Avoid MAOI's 14 days before starting venlafaxine and 7 days after stopping venlafaxine (5.2). Cimetidine: Caution in patients with pre-existing hypertension, in elderly patients and patients with hepatic dysfunction. (7.2) Haloperidol: Increase in Haloperidol AUC and C max . (7.4) Ketoconazole: Increase in venlafaxine and O-desmethylvenlafaxine AUC and C max . Caution when using venlafaxine with substances that inhibit both CYP2D6 and CYP3A4...."	

POSITIVE IMPACT	NEGATIVE IMPACT
Hyperlink: PubMed	Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Venlafaxine + Lithium Findings: Venlafaxine's DailyMed label states: "...Caution should be exercised with co-administration of venlafaxine and metoprolol. (7.8) CNS-active drugs: Caution when using venlafaxine with such drugs. (7.10) Serotonergic drugs (e.g., triptans, SSRIs, other SNRIs, linezolid, lithium, tramadol, or St. John's Wort): Potential for serotonin syndrome. Careful patient observation advised. (7.10) Tryptophan supplements: Concomitant use not recommended...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Venlafaxine + Paroxetine Findings: Paroxetine's DailyMed label states: "...Conversely, an increase in dosage of a CYP2D6 substrate may be needed if paroxetine is discontinued. Examples propafenone, flecainide, atomoxetine, desipramine, dextromethorphan, metoprolol, nebivolol, perphenazine, tolterodine, venlafaxine, risperidone...." Evidence: DailyMed label (Drug Interactions section); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed
	Venlafaxine + Quetiapine Findings: Quetiapine and Venlafaxine: Drug Interaction Risks Evidence: ClinicalTrials.gov; General web search (Tavily) Hyperlink: General web search (Tavily)
	Venlafaxine + Sertraline Findings: Sertraline's DailyMed label states: "...Conversely, an increase in dosage of a CYP2D6 substrate may be needed if sertraline hydrochloride is discontinued. Examples: propafenone, flecainide, atomoxetine, desipramine, dextromethorphan, metoprolol, nebivolol, perphenazine, thioridazine, tolterodine, venlafaxine Phenytoin Clinical Impact: Phenytoin is a narrow therapeutic index drug. Sertraline hydrochloride may increase phenytoin concentrations. Intervention: Monitor phenytoin levels when initiating or titrating sertraline hydrochloride. Reduce phenytoin dosage if needed...." Evidence: DailyMed label (Drug Interactions section); External evidence (PubMed); ClinicalTrials.gov; General web search (Tavily) Hyperlink: DailyMed

Known Limitations (Drug Interaction Flag Sheet):

- A pair with no finding from Flockhart, pharmacodynamic rules, PubMed, DailyMed, ClinicalTrials.gov, or Tavily is dropped from this report entirely, not shown as a negative/no-concern finding -- 'not listed here' means 'no evidence was found for this exact pair', never 'confirmed safe'.
- A DailyMed 'not mentioned in reviewed label section' result is never itself a finding -- it means this module checked the drug's real DRUG INTERACTIONS label section and the other drug's name was not there, not that no interaction exists; it only ever appears as supplementary context alongside whatever other evidence a pair has.
- Tavily (general web search) is queried directly and unconditionally alongside PubMed, DailyMed, and ClinicalTrials.gov for every pair here -- unlike elsewhere in this codebase (external_evidence_summary.py's question-answering paths), it is not gated behind the other sources returning no_results first. Every Tavily-sourced evidence entry is labeled 'General web search (Tavily)' so it is never mistaken for a dedicated medical/regulatory source, and it is always the lowest-priority candidate for a pair's primary Finding text.
- A source whose language does not clearly match an explicit positive-outcome or negative-outcome keyword never resolves a pair on its own -- this module falls through to the next candidate source in priority order instead, and if no candidate ever resolves clearly, the pair is dropped from the report entirely, the same as a pair with no evidence at all. There is no 'Needs Review' column here; most real titles/excerpts are topic descriptions, not verdicts, and this module never guesses which way an unclear one points.
- CPIC and ClinVar (gene-level sources) are not re-queried per pair here -- see the module docstring for why folding a single drug's own genotype-predicted finding into a pair's verdict here would conflate two different questions. Where this patient's genetics panel covers a screened drug, its real CPIC/ClinVar findings are in the Pharmacogenomics section, not duplicated here.
- Flockhart Cytochrome P450 Drug Interaction Table (curated-snapshot-2024) curated CYP-pathway data covers only a small hand-curated set of drugs; a pair outside that set relies on the other five sources, not a gap in this report.
- Positive/Negative classification is derived from keyword matching on publication titles, which can misread comparative-effectiveness study titles (where multiple drugs are studied together) as a verdict about one specific pair. This is a known limitation of title-only keyword classification, not a per-pair clinical conclusion.

Current Regimen — Curated-Source Coverage			
MEDICINE	STATUS	MEANING	ACTION
Lamotrigine	RESOLVED	Curated CYP/pharmacokinetic reference data exists for this drug (Flockhart Cytochrome P450 Drug Interaction Table).	No additional action required beyond the current-current pair evidence below.
Levetiracetam (Keppra)	RESOLVED	Curated CYP/pharmacokinetic reference data exists for this drug (Flockhart Cytochrome P450 Drug Interaction Table).	No additional action required beyond the current-current pair evidence below.

Clinical inference: Absence of a resolved current-current interaction does not establish absence of interaction. Unresolved predicted-model coverage limits interpretation; the active medication list should be reassessed against a complete interaction source before any treatment change. The therapy-level cards below give the supporting, counter, and unresolved evidence behind that conclusion for each current medication, across pharmacokinetic, pharmacodynamic, pharmacogenomic, clinical, EEG, and laboratory sources -- not a pass/fail interaction check.

- SuperCYPsPred predicted pharmacokinetic evidence is not available locally for any drug in this regimen.
- No therapeutic drug concentration data is present in the source reports.
- No renal function (eGFR) data is present in the source reports.

Candidate / Proposed Medication-Pair Screening

No candidate or proposed medications are present in this patient's regimen; no candidate-pair screening applies.

Lamotrigine **EFFECTIVENESS INCOMPLETE**

Therapeutic evidence present, but clinical effectiveness remains incomplete.

Supporting Evidence

- CBC status across the observation window: stable within reference ranges; no documented hematological toxicity. — digital_twin_state
- No documented drug toxicity for this agent in the clinical-notes source. — clinical_inference.medication_response
- SCN1A (CT): efficacy — findings_by_therapeutic_class

Counter-Evidence

- Breakthrough seizures persisted on the current regimen, with a recorded peak of 11 episodes on 2026-02-15. — clinical_inference.seizure_analysis
- ABCB1 (GG): reduced efficacy — findings_by_therapeutic_class
- CYP3A4 (TT): reduced efficacy — findings_by_therapeutic_class

Clinical Impression

4 supporting, 5 counter, and 13 unresolved evidence item(s) were identified for Lamotrigine. Clinical-notes inference: Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.

Recommended Monitoring

Continue EEG-derived seizure-risk monitoring.

Continue periodic CBC and safety-laboratory monitoring.

Continue seizure frequency, duration, and clustering tracking.

Review the priority safety flags in the Pharmacogenomics section before adding or changing an interacting medication.

Levetiracetam (Keppra) **EFFECTIVENESS INCOMPLETE**

Therapeutic evidence present, but clinical effectiveness remains incomplete.

Supporting Evidence

- CBC status across the observation window: stable within reference ranges; no documented hematological toxicity. — digital_twin_state
- No documented drug toxicity for this agent in the clinical-notes source. — clinical_inference.medication_response
- SCN1A (CT): efficacy — findings_by_therapeutic_class

Counter-Evidence

- Breakthrough seizures persisted on the current regimen, with a recorded peak of 11 episodes on 2026-02-15. — clinical_inference.seizure_analysis
- Seizure clustering was documented during the observation window. — clinical_inference.seizure_analysis
- EEG synthesis remained consistent with active epilepsy (EEG-derived risk: Elevated; future seizure probability: Elevated). — digital_twin_state

Clinical Impression

3 supporting, 3 counter, and 7 unresolved evidence item(s) were identified for Levetiracetam (Keppra). Clinical-notes inference: Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.

Recommended Monitoring

Continue EEG-derived seizure-risk monitoring.

Continue periodic CBC and safety-laboratory monitoring.

Continue seizure frequency, duration, and clustering tracking.

Review the priority safety flags in the Pharmacogenomics section before adding or changing an interacting medication.

Limitations of this screen: This assessment is clinical decision support, not an automated prescribing system; no dose, start, or stop decision is made or implied here. No proposed/candidate medication list is present in the source data, so only the current regimen's pairwise and therapy-level evidence is evaluated. Curated pharmacokinetic (CYP) coverage is limited to a small bundled reference table; a drug not listed there is reported as unresolved, not as free of interactions. Therapy assessments also draw on the live external-evidence layer (see Known Limitations under Pharmacogenomics, above, for its source-by-source scope and caveats).

07 Pharmacogenomics

76 variants across 47 medications · reported 10 Sep 2026

Pharmacogenomic Panel Summary

CYP2D6 GG	CYP2C19 GA	CYP2C9 *1/*3 - Intermediate Metabolizer	CYP2B6 *1/*6 - Intermediate Metabolizer	CYP3A5 *3/*3 - Poor Metabolizer
CYP1A2 AA	CYP3A4 TT	UGT1A4 *1/*3A - Decreased Function	POR *1/*28 - Decreased Function	

Interpretive method: a pharmacogenomic association is weighed against this patient's own observed clinical response, laboratory, and EEG evidence -- it is never used as an isolated prescribing instruction, and a favorable genotype cannot override documented toxicity or persistent symptoms. The table below shows the single most clinically significant finding per current-regimen drug; the full therapy-level assessment -- this finding combined with clinical, EEG, laboratory, and external-database evidence -- is presented in Section 06, Drug Interactions.

Patient-Specific PGx Findings

Therapy	Gene/Genotype	Patient-Specific Interpretation	Clinical Relevance
Lamotrigine	CYP3A4 (TT)	Genotype is associated with a reduced-efficacy response to this therapy.	Moderate relevance; flagged as clinically significant in the source panel (citation: PMID:28343093).
Levetiracetam	SCN1A (CT)	Genotype is associated with a favorable efficacy response to this therapy.	Supportive relevance; flagged as clinically significant in the source panel (citation: PMID:28753467).

External Evidence Verification

Database evidence only, not patient-verified: each block below is a live public-database lookup for the same gene/drug pair as the patient-specific finding in Table 1 above -- it establishes general scientific plausibility for that pair, not a fact confirmed for this individual patient. An unresolved result means this exact query returned nothing, or failed, at the time checked -- it is not evidence that no relevant data exists, and must not be read as a clean or reassuring negative.

- CYP3A4 → Lamotrigine** PARTIALLY VERIFIED
 - CPIC: Unresolved. ClinVar: Uncertain significance.
 - PubMed: PMID 21635243 found. Overall: Partially resolved (2 of 3 sources).
- SCN1A → Levetiracetam** PARTIALLY VERIFIED
 - CPIC: Unresolved. ClinVar: Uncertain significance.
 - PubMed: PMID 39200163 found. Overall: Partially resolved (2 of 3 sources).

Provenance

- CYP3A4 → Lamotrigine:** ClinVar — Accession VCV004885026 (70 total gene-wide records) · LIVE · Retrieved 11 Sep 2026; PubMed — PMID 21635243 · LIVE · Retrieved 11 Sep 2026. Note: Table 1's citation is from the patient's own PGx report; this is an independent live PubMed search, may differ.
- SCN1A → Levetiracetam:** ClinVar — Accession VCV004887952 (5406 total gene-wide records) · LIVE · Retrieved 11 Sep 2026; PubMed — PMID 39200163 · LIVE · Retrieved 11 Sep 2026. Note: Table 1's citation is from the patient's own PGx report; this is an independent live PubMed search, may differ.

Evidence Scope: live-verified via ClinVar, CPIC, DailyMed, PubMed, and ClinicalTrials.gov only. The rest of the NeuroTwin resource catalog (PharmGKB, DrugBank, Reactome, STRING, GTEx, and others) is out of scope for this live-lookup layer and is not reflected anywhere in this table — see Known Limitations below.

Known Limitations (External Evidence):

- This module queries exactly five public APIs (PubMed, ClinVar, DailyMed, CPIC, ClinicalTrials.gov); it does not cover the rest of the NeuroTwin resource catalog (Reactome, STRING, GTEx, RCSB PDB, etc.) -- those remain out of scope for this module and must not be inferred from it.
- A 'no_results' status means the API returned zero matching records for the exact query issued; it is not evidence of absence, and must not be reported as a negative finding.
- Responses may be served from the on-disk cache in reports/external_evidence/cache/ rather than a fresh call; each record's 'from_cache' and 'retrieved_at' fields state which. Pass --refresh to force live calls.

35
Total Recordings

12
Interictal

11
Preictal

12
Ictal

Signal Variance Midpoint by Phase (µV²)

344.45

Interictal

4343.59

Preictal

42395.2

Ictal

EEG-derived risk: Elevated

Future seizure probability: Elevated

Dominant feature: Signal Variance

Phase Analysis

PHASE	STATE	ACTIVITY	VARIABILITY
Interictal	Baseline	Stable	Low
Preictal	Transition	Increasing Instability	Moderate-High
Ictal	Seizure	Hypersynchronous	Very High

PHASE	AMPLITUDE RANGE	STD DEV RANGE	VARIANCE RANGE
Interictal	-89; 99	11.68; 23.5	136.44; 552.46
Preictal	-345; 440	14.48; 92.07	209.71; 8477.48
Ictal	-504; 784	52.5; 286.42	2756.47; 82033.96

Longitudinal EEG recordings demonstrate a consistent transition from stable interictal activity to preictal electrical instability followed by high-amplitude ictal seizure activity. Findings are consistent with the dataset's interictal/preictal/ictal phase taxonomy, indicating reproducible seizure-related signal evolution; this is a signal-classification observation, not a clinician-verified diagnosis.

09MRI / CT

Structural imaging: neuroimaging key findings

CT — 6 studies, 12 Jan – 29 Jun 2026

DATE	IMPRESSION
12 Jan 2026	No acute intracranial abnormality is evident on this single-image model-assisted review.
16 Feb 2026	No gross acute abnormality is evident on this limited skull-base level image; assessment is limited by slice level and image quality.
11 Mar 2026	No acute intracranial abnormality is evident on this single-image model-assisted review.
20 Apr 2026	No acute intracranial abnormality is evident on this high-convexity image; ventricular assessment is not available at this slice level.
14 May 2026	No acute intracranial abnormality is evident on this limited-quality near-vertex image.
29 Jun 2026	No acute intracranial abnormality is evident. Incidental bilateral choroid plexus calcifications are present.

Across the available CT image files, no gross acute intracranial hemorrhage, midline shift, or mass effect is evident on this model-assisted image-only review. Direct progression assessment is limited because the files represent single slices with different slice levels, resolutions, and image quality. This output is not a radiologist-verified diagnostic report.

MRI — Metadata-supported structural stability, diagnostic interpretation not available

FINDING	STATUS	DETAIL
Brain Morphology	NOT DIRECTLY ASSESSED	The provided files include volumetric MRI metadata, but no radiologist interpretation or morphometric segmentation was available for direct morphology assessment.
Brain Volume	STABLE WITHIN AVAILABLE METADATA	T1w nonzero volume range: 8387642 to 8777268 mm3 (mean 8641065 mm3; range 4.5%). FLAIR nonzero volume range: 6022385 to 6625690 mm3 (mean 6231344 mm3; range 9.7%). These are preprocessing-derived nonzero voxel volumes, not formal brain volumetry.
Tissue Signal Characteristics	QUANTITATIVELY SUMMARIZED, NOT DIAGNOSTICALLY INTERPRETED	FLAIR mean intensity 28.76 to 32.27; standard deviation 47.57 to 54.16; T1w mean intensity 99.82 to 105.56; standard deviation 153.13 to 162.05. Intensity statistics are useful for preprocessing consistency but are not diagnostic tissue-signal interpretation.

Not directly assessed from the available imaging metadata (this means "not evaluated," not "abnormal"): White Matter, Hippocampus, Cortical Structure, Mass Lesion, Hemorrhage, Infarction.

Stable by available preprocessing metadata. 6 MRI series were available across 3 study identifiers, including 3D T1-weighted MRI, T2-FLAIR MRI. Preprocessed metadata show consistent 3D acquisition characteristics and no metadata-level evidence of major global structural progression. A radiology report, lesion segmentation, hippocampal volumetry, and cortical thickness analysis were not available, so subtle pathology cannot be excluded from these metadata alone.

Generated from validated patient data. Decision-support summary only -- not a substitute for clinician review.

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10 ECG

single channel ("ECG SD") in every recording

12

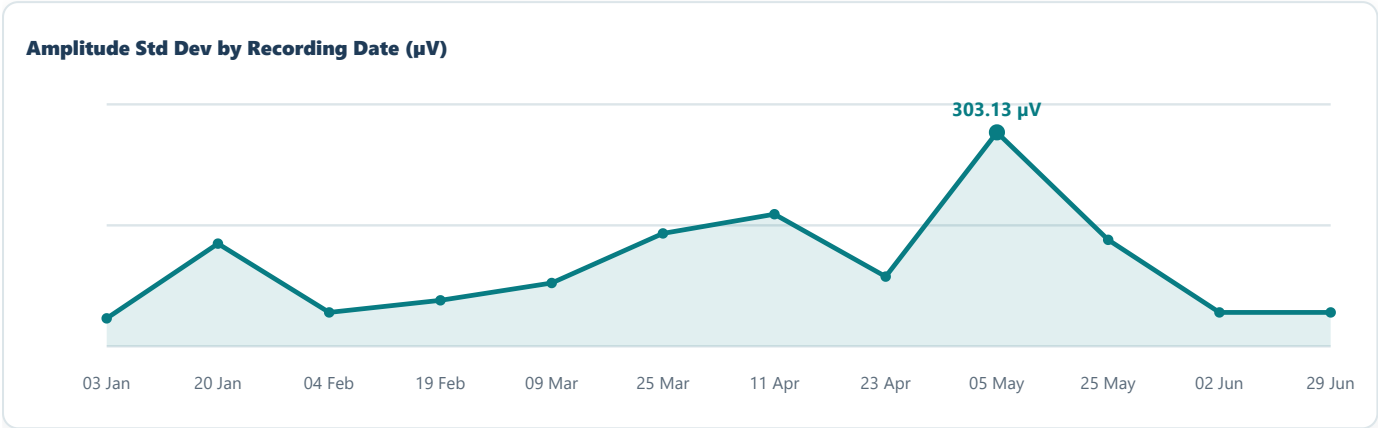
Recordings

99h 11m

Total Duration

256

Sampling Rate (Hz)



- All 12 recordings use a single channel sampled at 256 Hz, so statistics are directly comparable across sessions.
 - Recording duration varies considerably, from about 1h 11m (2026-04-11) to about 22h 27m (2026-03-25).
 - The 2026-06-02 and 2026-06-29 recordings have matching percentile values and near-identical histogram shapes, suggesting very similar signal characteristics or a data duplication between the two sessions.
 - Standard deviation of amplitude ranges from about 82.6 uV (2026-01-03) to about 303.1 uV (2026-05-05), indicating substantially different signal variability/noise levels between sessions.
 - Peak-to-peak amplitude (max minus min) ranges from roughly 2,483 uV (2026-05-25) to about 3,577 uV (2026-03-25).
- This report is derived strictly from the statistical fields present in the source dataset. It does not represent a clinical or diagnostic ECG interpretation. Any assessment of cardiac rhythm, heart rate, or abnormality requires review of the underlying waveform by a qualified clinician or an appropriately validated diagnostic algorithm, neither of which is reflected in this report.

11 CBC / Labs

12 reports, 01 Jan – 30 Jun 2026

Overall hematological status: Stable within reference ranges. No CBC parameter abnormality was identified against the report-provided reference ranges across the observation period.

PARAMETER	FIRST	LATEST	RANGE	TREND
Hemoglobin (Hb)	14.4	15.1	13.5 – 15.9	increasing
Total RBC count	5.0	5.1	4.8 – 5.3	increasing
Packed Cell Volume (PCV)	44.7	43.5	42.2 – 48.4	decreasing
Mean Corpuscular Volume (MCV)	90.0	97.8	88.4 – 98.8	increasing
MCH	27.6	29.9	27.6 – 31.1	increasing
MCHC	33.1	33.0	32.9 – 34.3	decreasing
RDW	12.1	13.5	12.1 – 13.5	increasing
Total WBC count	5761.0	9630.0	5476.0 – 9973.0	increasing
Neutrophils	58.0	54.0	52.0 – 61.0	decreasing
Lymphocytes	22.0	29.0	22.0 – 34.0	increasing
Eosinophils	4.0	4.0	1.0 – 5.0	stable
Monocytes	3.0	3.0	1.0 – 8.0	stable
Basophils	1.0	1.0	0.0 – 2.0	stable
Platelet Count	251725.0	209427.0	189551.0 – 373102.0	decreasing

Patient-reported vs. measured discrepancy: the 30 Jun 2026 questionnaire records patient recollections of White Blood Cell (WBC) count baseline trend from your latest Complete Blood Count (CBC): Mild Leukopenia (Borderline low); Platelet Count profile stability from recent blood evaluations: Borderline Thrombocytopenia (Low); Liver Function Test (LFT) - SGOT/AST status: Mild Elevation (40-100 U/L); measured CBC values across all 12 reports in this period remained within their reference ranges throughout (see table below). The discrepancy likely reflects patient recall/perception rather than a measured lab abnormality.

12

Sleep, Cognition & Quality of Life

Sleep		Cognition		Quality of Life	
TREND	Sleep remained variable across visits	MEMORY	Stable on clinician review, with low to intermittent patient-reported memory difficulty	DAILY FUNCTION	Independence in daily activities was repeatedly normal, intact, or stable
AVERAGE	6.2 hours per night across printed assessments	ATTENTION	Variable patient-reported concentration and attention symptoms	SOCIAL FUNCTION	Intermittent social withdrawal or isolation symptoms were reported
Daytime naps needed		LANGUAGE	Mild word-finding difficulty was noted but not described as worsening	CLINICAL INTERPRETATION	
Difficulty falling back asleep		MENTAL CLARITY	Improved mental clarity was reported at the latest visit	Quality of life remains affected by seizures, sleep, triggers, and medication tolerability, despite generally stable outpatient status.	
Fatigue level through the day					
Waking up unrefreshed					

Latest Questionnaire Domain Scores (0-100)		
Sleep Quality & Architecture		91.7/100
Cognitive & Executive Function		83.3/100
Lab Biomarkers & Lifestyle		75/100

13

Digital Twin Risk Dashboard

Overall clinical risk: Moderate

Future Seizure Risk Moderate	Status Epilepticus Risk Low To Moderate	Hospitalization Risk Low To Moderate	Cognitive Decline Risk Low
Medication Toxicity Risk Low To Moderate	Functional Decline Risk Low	EEG-Derived Seizure Risk Elevated	Genetic Safety Flags 10 Flags, 7 High

State Vector (Health Metric Scores, 0-100)		
Disease Stability		67/100
Treatment Response		64/100
Medication Safety		67/100
Cognitive Health		69/100
Sleep Health		66/100
Quality Of Life		67/100
Each domain score is the mean of clinician-documented per-visit ratings across the clinical-note window; overall health score is the mean of the other domains. Derived digital-twin summary metrics, not a direct lab/imaging measurement.		

14 Clinical Action Plan

Priority Actions (Pharmacogenomics)

- HIGH** **Avoid or use with heightened monitoring**
See priority_safety_flags for the specific gene-drug pairs (ABCB1 (CC), BDNF (TC), CREB1 (AA), FKBP5 (TC), GRIA3 (A), HTR1B (TA), METTL21A (AA)).
- MODERATE** **Dose with caution / consider therapeutic drug monitoring**
See metabolizer_profile for reduced-function genes (CYP2C9, CYP2B6, CYP3A5, UGT1A4, POR).
- INFORMATIONAL** **Contextualize**
Interpret alongside actual clinical history, current medication list, and labs -- this panel alone does not establish a diagnosis.

Recommended Actions

- continue the documented antiseizure medication regimen unless changed by the treating clinician
- maintain the documented plan for consistent sleep schedule and trigger avoidance
- continue seizure and aura tracking across follow-up windows
- continue periodic CBC, liver function, and antiseizure drug-level safety monitoring when scheduled
- reassess at the planned follow-up interval

Monitoring Priorities

- seizure frequency by type
- seizure clustering
- average and longest seizure duration
- post-ictal recovery time
- aura pattern and severity
- sleep duration and sleep fragmentation
- missed medication doses and trigger exposure
- dizziness and other medication side effects
- CBC, liver function, and antiseizure drug-level tracking
- word-finding, memory, and attention symptoms
- Signal Variance
- Amplitude Dynamics
- Electrical Stability
- Preictal Detection
- Seizure Frequency
- Temporal Disease Progression

MRI Follow-Up

- Perform formal radiology review of the MRI image volumes.
- Run automated brain segmentation and longitudinal registration across the three study identifiers.
- Perform hippocampal volumetry because epilepsy-related mesial temporal abnormalities cannot be excluded from metadata alone.
- Run FLAIR white matter lesion analysis if clinically relevant.
- Compare future MRI studies using the same T1w and FLAIR acquisition strategy when possible.
- Correlate MRI findings with seizure semiology and neurological examination.
- Review MRI together with EEG findings for seizure-focus localization.
- Repeat or escalate imaging review if new neurological deficits, worsening seizures, or new focal symptoms occur.

Medication Reconciliation

- Current regimen reconciled and checked against curated reference pharmacology and this patient's own pharmacogenomic findings (Section 07). Predicted-model (SuperCYPsPred) coverage is still unresolved, so a live, comprehensive DDI screen is recommended before adding or changing any interacting medication.